

IEOR E4004

Optimization Models and Methods

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Outline

- Introduction to Linear Programming (LP)
 - Formulating LPs
- Solving Linear Programs
 - Graphical Solution Method
- Summary

Example Formulation Problem

A Disaster Aid Problem

A Disaster Aid Problem



After a major disaster, an aid organization wants to quickly send tents and food using a cargo plane.

Maximize the combined benefit from tents and food sent:

- The total weight of **all pallets** cannot exceed **40,000 lbs**
- There is space for at most **6 pallets** on a plane
- Only up to **5 pallets of food** are currently available to send
- We **cannot send a negative number** of pallets
- The benefits are **8 units per each pallet of tent** and **6 units per each pallet of food** sent
- The following table lists the weight, cost, criticality, expected benefit of each pallet.

A Disaster Aid Problem



Table: Data for Sending Aid

	Tents	Food
Weight (1000 lbs)	7.5	5
Cost (\$1000/pallet)	11	4
Criticality	5	8
Expected Benefit	8	6

**How many pallets of tents and food must be sent
to achieve the highest possible benefit?**

A Disaster Aid Problem: Formulation



Step 1: Define decision variables

01



How many pallets of tent should we send in a cargo plane?



How many pallets of food should we send in a cargo plane?

A Disaster Aid Problem: Formulation



Step 2: Form objective function

What is our aim/objective?

To find the loading plan that will **maximize** the benefit

$$\text{Benefit: } 8 \text{  + 6 \text{ $$

02

$$\textit{maximize} \quad 8 \text{  + 6 \text{ $$

A Disaster Aid Problem: Formulation



Step 3: Add constraints

x



produced each week

y



produced each week

03

subject to

$7.5x + 5y \leq 40$	<i>Weight capacity</i>
$x + y \leq 6$	<i>Space limit</i>
$y \leq 5$	<i>Food available</i>
$x, y \geq 0$	<i>Non-negativity clause</i>

A Disaster Aid Problem: Formulation



Step 4: Complete problem

max $8x + 6y$ *Objective function*

s.t $7.5x + 5y \leq 40$ *Constraints*

$$x + y \leq 6$$

$$y \leq 5$$

$$x, y \geq 0$$

Example Formulation

Linear Regression

Recall Linear Regression

We are given vectors (x_i, y_i) where $i \in \{1, \dots, N\}$

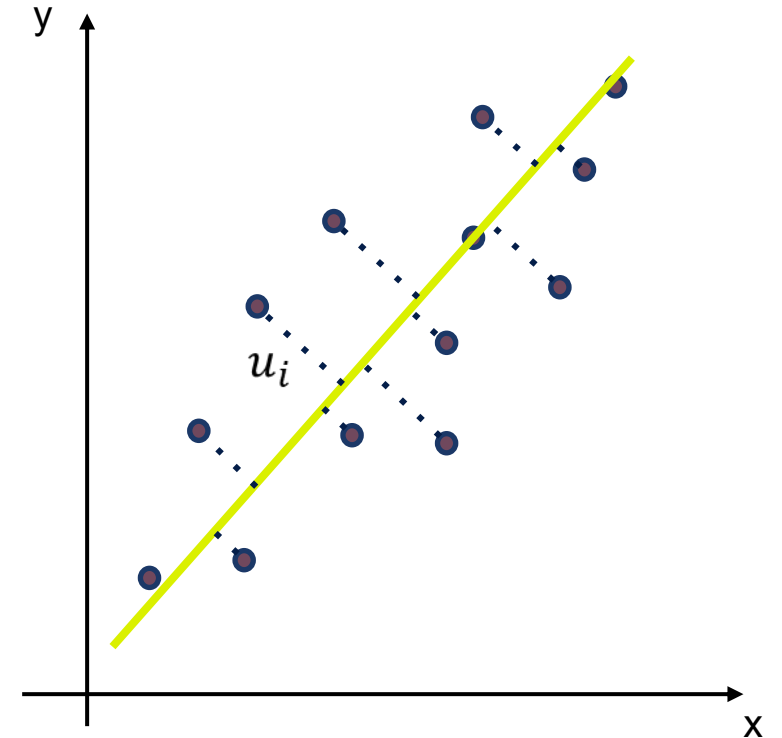
We want to “approximate” a linear model for the data

We want to find vectors $w \in \mathbb{R}^n, z \in \mathbb{R}$, so that:

$$y_i \approx w^T x_i + z, \quad 1 \leq i \leq N$$

For a given choice of w, z , the error on (x_i, y_i) is:

$$u_i = y_i - (w^T x_i + z)$$



Recall Linear Regression

We are given vectors

We are given scalar values

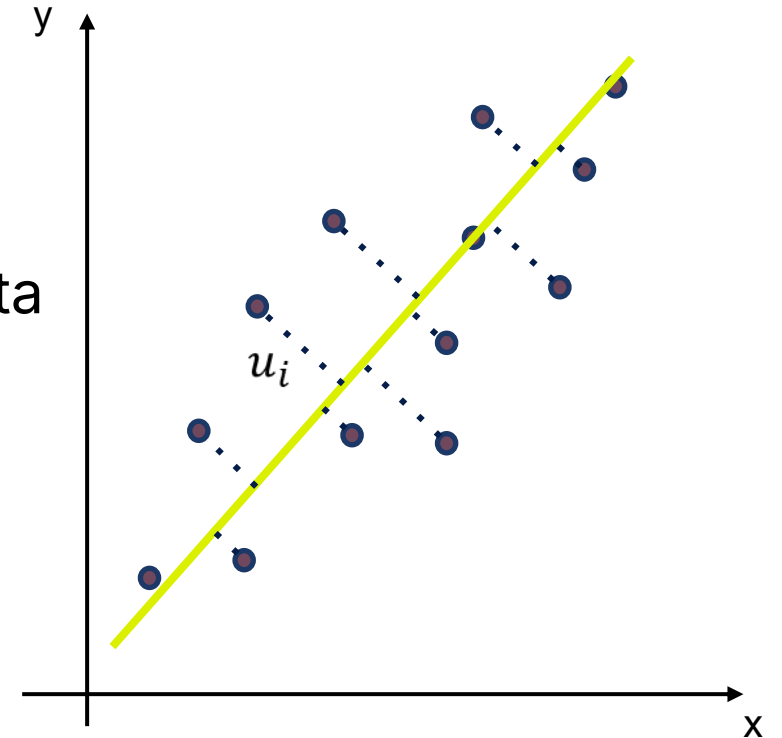
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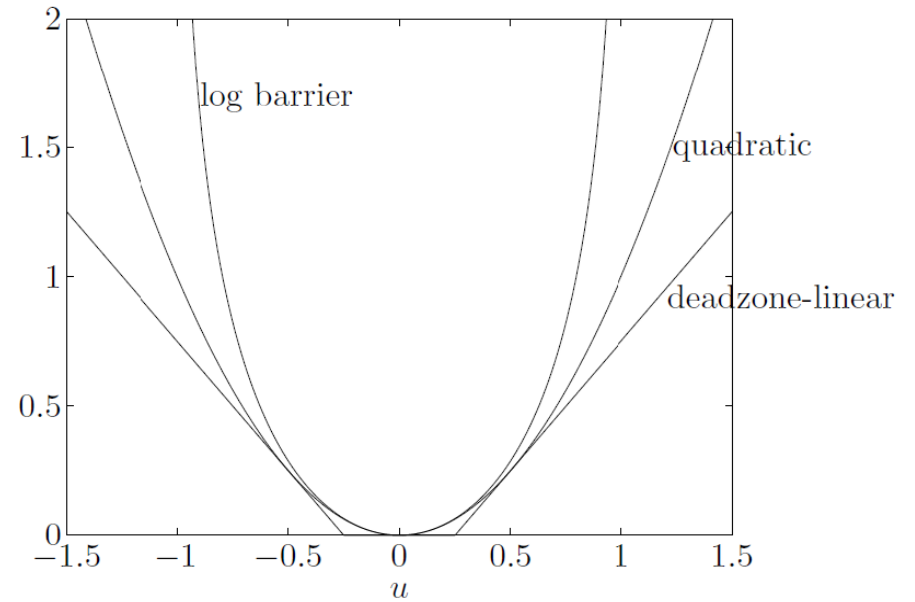


A linear regression model solves $\min_{w,z} \ell(u)$ for an appropriate function ℓ

Linear Regression Models

Some common choices for l :

- **Quadratic** loss: $l(u) = \sum_{i=1}^N u_i^2 =: u^2$
- **Absolute** loss or L_1 loss: $l(u) = \sum_i |u_i| = |u|$
- **Deadzone** linear: $l(u) = \begin{cases} 0, & |u| < 1 \\ 2|u| - 1, & \text{otherwise} \end{cases}$
- **Huber** penalty: $l(u) = \begin{cases} u^2, & |u| < 1 \\ 2|u| - 1, & \text{otherwise} \end{cases}$
- **Log-barrier**: $l(u) = -\log(1 - \frac{u^2}{M})$ for some constant M



Example: Linear Regression

How can we find the best linear relationship between the housing price and the housing size?

Let:

- y = House price in \$1000s
- x_1 = Size (sq ft)

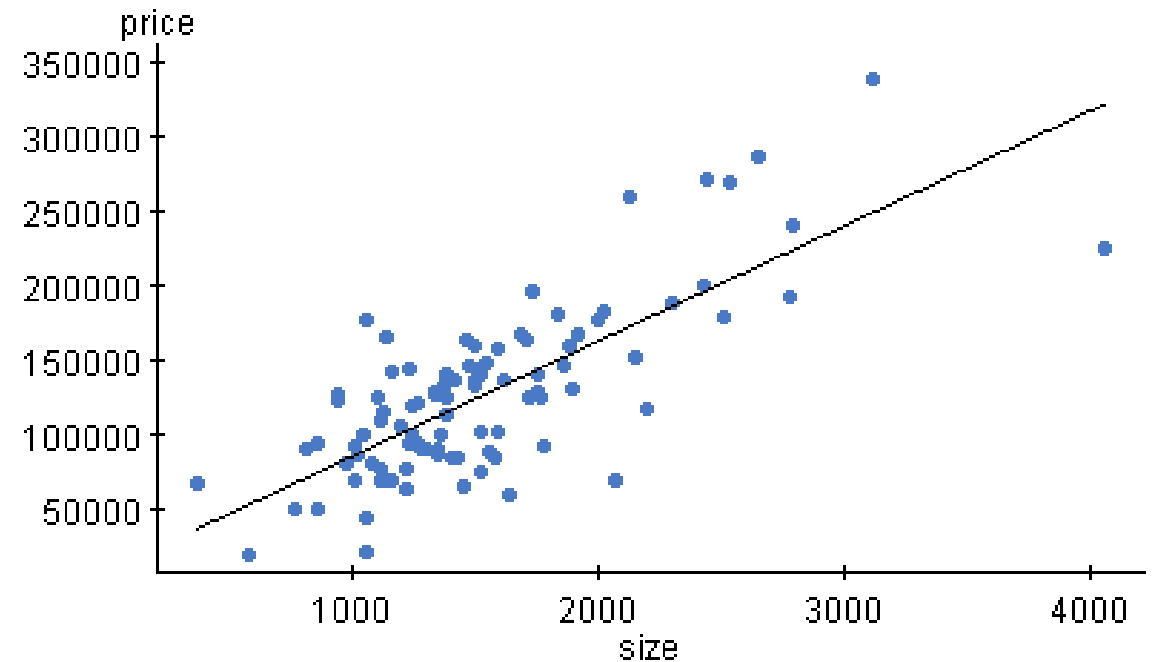


Figure: Example Graph Showcasing Potential Relationship Between Housing Size and Price

Example: Linear Regression

Let:

- y = House price in \$1000s
- x_1 = Size (sq ft)

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \quad \text{for } i = 1, \dots, n$$

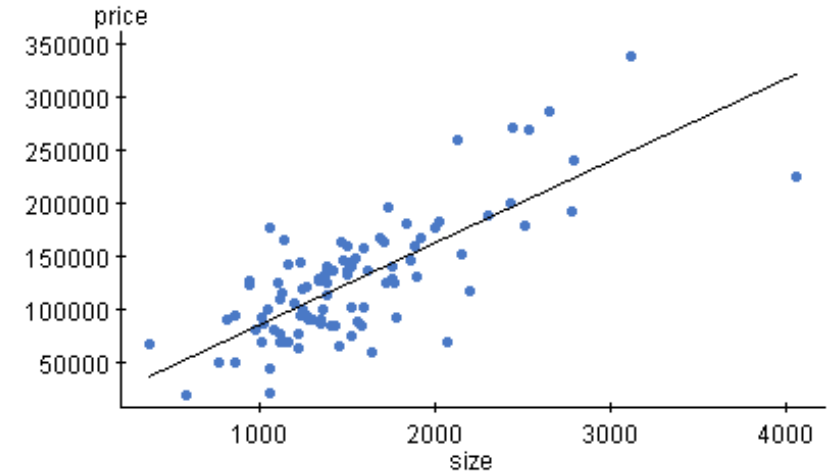


Figure: Example Graph Showcasing Potential Relationship Between Housing Size and Price

Using Least Absolute Deviations we minimize **absolute errors**:

$$\min_{\beta_0, \beta_1} \sum_{i=1}^n |\varepsilon_i| = \sum_{i=1}^n |y_i - (\beta_0 + \beta_1 x_i)|$$

Example: Linear Regression

Using **Least Absolute Deviations** we minimize **absolute errors**:

Decision variables

- $\beta_0 \in \mathbb{R}$ (intercept)
- $\beta_1 \in \mathbb{R}$ (slope)
- $u_i \geq 0$ for $i = 1, \dots, n$ (absolute residuals)

$$\min_{\beta_0, \beta_1} \sum_{i=1}^n |\varepsilon_i| = \sum_{i=1}^n |y_i - (\beta_0 + \beta_1 x_i)|$$

We can also linearize this by:

$$\begin{aligned} \min_{\beta_0, \beta_1, u} \quad & \sum_{i=1}^n u_i \\ \text{s.t.} \quad & y_i - (\beta_0 + \beta_1 x_i) \leq u_i \quad \forall i \\ & (\beta_0 + \beta_1 x_i) - y_i \leq u_i \quad \forall i \\ & u_i \geq 0 \quad \forall i \end{aligned}$$

Introduction to Linear Programming

Linear Programming

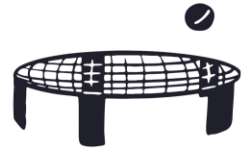
- A **linear programming (LP)** problem is an optimization problem where:
 - We try to maximize (or minimize) a **linear function** of **continuous** decision variables.
 - Objective function: The function that is to be maximized or minimized
 - Decision variables are allowed to take fractional values

Topics

Formulating Linear Programmings (LPs):

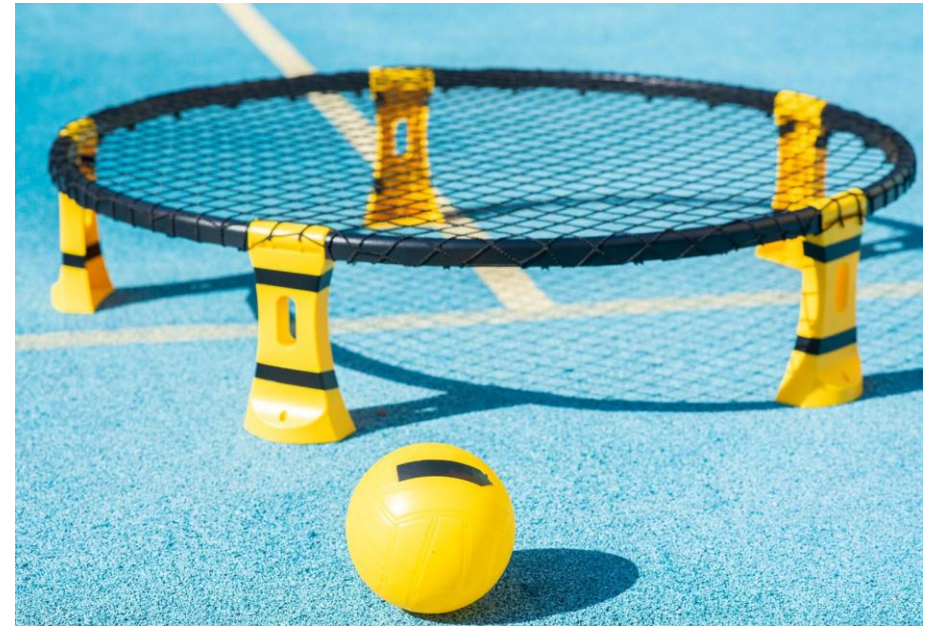
- A Resource Allocation Problem
- A Purchasing Problem

A Resource Allocation Problem

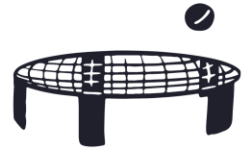


SpikePlay Co. produces popular outdoor sports gear:

- **Spikeball Set:** 1 net + 1 ball + packaging
- **Beach Spikeball Set:** Compact version, lighter net
- **Individual Ball:** Sold as replacement or for extra practice



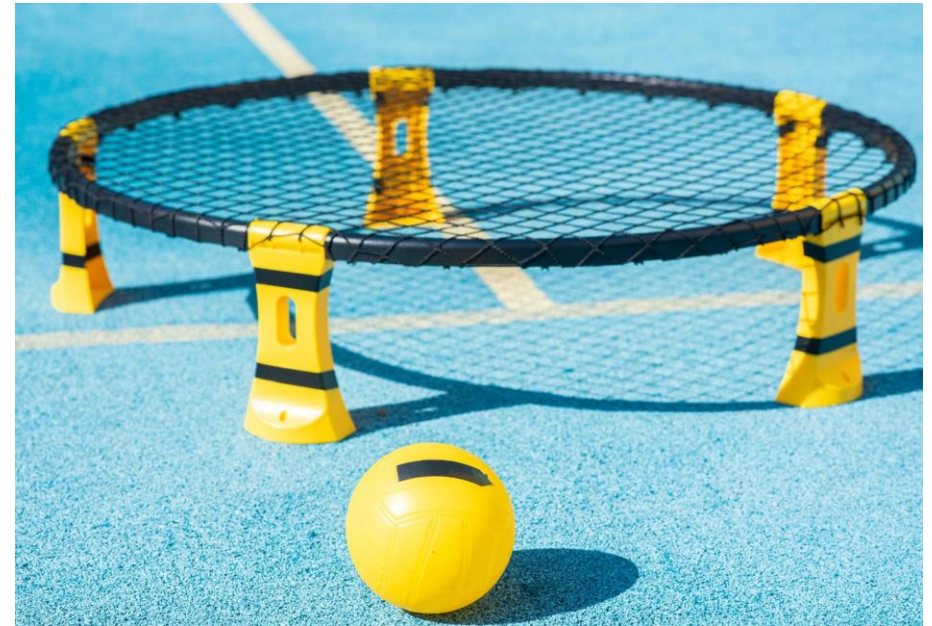
A Resource Allocation Problem



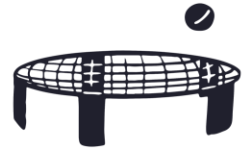
SpikePlay Co. produces popular outdoor sports gear:

- **Spikeball Set:** 1 net + 1 ball + packaging
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- **Individual Ball:** Sold as replacement or for extra practice

Due to supply chain delays, materials are limited this month. SpikePlay wants to **plan production to maximize profit** using on-hand inventory.



A Resource Allocation Problem



Material / Info	Classic Set	Beach Set	Extra Ball	Available
Plastic (oz)	48	40	0	240,000 oz
Balls (units)	1	1	1	8,000 units
Packaging (ft ²)	10	10	2	75,000 ft ²
Selling Price (\$)	40	38	9	—
Profit Margin (\$)	10	12	2	—

The forecasted demand for the Beach Set at this price is 2000 for the month

A Resource Allocation Problem | Decision Variables

Step 1: Set decision variables

A Resource Allocation Problem | Decision Variables

Step 1: Set decision variables



What is the decision SpikePlay Co. wants to make?

A Resource Allocation Problem | Decision Variables

Step 1: Set decision variables



What is the decision SpikePlay Co. wants to make?

How many of each product type to produce?

How much of each resource to allocate to production?

A Resource Allocation Problem | Decision Variables

Step 1: Set decision variables



What is the decision SpikePlay Co. wants to make?

Therefore,

Let C , B , and E denote the number of Classic Sets, Beach Sets and Extra Balls produced this month, respectively.

A Resource Allocation Problem | Objective

Step 2: Set objective

A Resource Allocation Problem | Objective

Step 2: Set objective

? What is the goal SpikePlay Co. wants to achieve?

A Resource Allocation Problem | Objective

Step 2: Set objective

? What is the goal SpikePlay Co. wants to achieve?

Maximizing their monthly profit with a new production plan

A Resource Allocation Problem | Objective

Step 2: Set objective

? What is the goal SpikePlay Co. wants to achieve?

We can express the total monthly profit as:

$$\textit{Maximize} \quad 10C + 12B + 2E$$

A Resource Allocation Problem | Constraints

Step 3: Set constraints

A Resource Allocation Problem | Constraints

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What are the limitations SpikePlay Co. have?

A Resource Allocation Problem | Constraints

Step 3: Set constraints



What are the limitations SpikePlay Co. have?

- **The total amount of materials (plastic, balls, and the packaging) used must be less than or equal to the amount of inventory available**

A Resource Allocation Problem | Constraints

Step 3: Set constraints



What are the limitations SpikePlay Co. have?

- **The total amount of materials (plastic, balls, and the packaging) used must be less than or equal to the amount of inventory available**
- **The forecasted upper bound on the Beach Set demand should not be exceeded**

A Resource Allocation Problem | Constraints

- The total amount of materials (plastic, balls, and the packaging) used must be less than or equal to the amount of inventory available

$$48C + 40B \leq 240,000 \quad (\text{Plastic})$$

$$C + B + E \leq 8,000 \quad (\text{Ball})$$

$$10C + 10B + 2E \leq 75,000 \quad (\text{Packaging})$$

A Resource Allocation Problem | Constraints

- The forecasted upper bound on the Beach Set demand should not be exceeded

$$B \leq 2,000 \quad (\text{Demand for the Beach Set})$$

The Complete LP

Maximize $10C + 12B + 2E$

Subject to $48C + 40B \leq 240,000$ (Plastic)

$C + B + E \leq 8,000$ (Ball)

$10C + 10B + 2E \leq 75,000$ (Packaging)

$B \leq 2,000$ (Demand for the Beach Set)

$C, B, E \geq 0$

A Purchasing Problem



AutoPartsNow is an auto parts retailer launching an online ad campaign using two platforms:

- **Web Ads:** Banner & popup ads on websites
- **Social Media:** Ads in user newsfeeds



A Purchasing Problem



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Their aim is to reach potential customers aged **25-35** and **36+**, while **minimizing the total ad costs**.

A Purchasing Problem



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- **Web Ads:** Banner & popup ads on websites
- **Social Media:** Ads in user newsfeeds

Their aim is to reach potential customers aged **25-35** and **36+**, while **minimizing the total ad costs**.

They are only required to pay **per click**, and want to:

- Hit **click targets** for both age groups

A Purchasing Problem



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Their aim is to reach potential customers aged **25-35** and **36+**, while **minimizing the total ad costs**.

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- Hit **click targets** for both age groups
- Ensure at least **50% of all visitors are unique**

A Purchasing Problem



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- **Social Media:** Ads in user newsfeeds

Their aim is to reach potential customers aged **25-35** and **36+**, while **minimizing the total ad costs**.

They are only required to pay **per click**, and want to:

- Hit **click targets** for both age groups
- Ensure at least **50% of all visitors are unique**
- Don't exceed the maximum click limit

A Purchasing Problem



	Banner	Popup	Newsfeed
Cost (\$/1,000 clicks)	65	110	120
Clicks (25-35)	35%	30%	70%
Clicks (36+)	60%	60%	30%
Unique Visitors	40%	70%	80%
Max Clicks (×1000)	600	300	300

Age Group	Target Clicks
(25-35)	500,000
36+	600,000

A Purchasing Problem | Decision Variables

Step 1: Set decision variables

A Purchasing Problem | Decision Variables

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What is the decision AutoPartsNow wants to make?

A Purchasing Problem | Decision Variables

Step 1: Set decision variables



What is the decision AutoPartsNow wants to make?

How much they should invest on different types of advertising campaigns?

A Purchasing Problem | Decision Variables

Step 1: Set decision variables

? What is the decision AutoPartsNow wants to make?

How much they should invest on different types of advertising campaigns?

Therefore, let x_i be the number of clicks (in 1000s) on type i ads, for $i = 1$ (*banner*), 2 (*popup*), 3 (*news feed*).

A Purchasing Problem | Objective

Step 2: Set the objective function

A Purchasing Problem | Objective

Step 2: Set the objective function



What does AutoPartsNow wants to achieve?

A Purchasing Problem | Objective

Step 2: Set the objective function



What does AutoPartsNow wants to achieve?

They aim to reach potential customers while minimizing the amount they spend

A Purchasing Problem | Objective

Step 2: Set the objective function



What does AutoPartsNow wants to achieve?

They aim to reach potential customers while minimizing the amount they spend

Therefore, the objective function will be:

$$\text{minimize } 65x_1 + 110x_2 + 120x_3$$

A Purchasing Problem | Constraints

Step 3: Set constraints

A Purchasing Problem | Constraints

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What are the limitations AutoPartsNow have?

A Purchasing Problem | Constraints

Step 3: Set constraints



What are the limitations AutoPartsNow have?

- The total clicks from age 25-35 visitors must be at least 500,000

A Purchasing Problem | Constraints

Step 3: Set constraints



What are the limitations AutoPartsNow have?

- The total clicks from age 25-35 visitors must be at least 500,000
- The unique click from age 25-35 visitors must be at least 50% of their total clicks

A Purchasing Problem | Constraints

Step 3: Set constraints



What are the limitations AutoPartsNow have?

- The total clicks from age 25-35 visitors must be at least 500,000
- The unique click from age 25-35 visitors must be at least 50% of their total clicks
- The total clicks from older visitors must be at least 600,000

A Purchasing Problem | Constraints

Step 3: Set constraints



What are the limitations AutoPartsNow have?

- The total clicks from age 25-35 visitors must be at least 500,000
- The unique click from age 25-35 visitors must be at least 50% of their total clicks
- The total clicks from older visitors must be at least 600,000
- The unique clicks from older visitors must be at least 50% of their total clicks

A Purchasing Problem | Constraints

Step 3: Set constraints



What are the limitations AutoPartsNow have?

- The total clicks from age 25-35 visitors must be at least 500,000
- The unique click from age 25-35 visitors must be at least 50% of their total clicks
- The total clicks from older visitors must be at least 600,000
- The unique clicks from older visitors must be at least 50% of their total clicks
- The maximum click bound should not be exceeded

A Purchasing Problem | Constraints

- The total clicks from **age 25-35 visitors** must be at least 500,000

$$0.35x_1 + 0.3x_2 + 0.7x_3 \geq 500$$

- The unique click from **age 25-35 visitors** must be at least 50% of their total clicks (250,000)

$$0.4(0.35x_1) + 0.7(0.3x_2) + 0.8(0.7x_3) \geq 0.5(500)$$

A Purchasing Problem | Constraints

- The total clicks from **older visitors** must be at least 600,000

$$0.6x_1 + 0.6x_2 + 0.3x_3 \geq 600$$

- The unique clicks from **older visitors** must be at least 50% of their total clicks (300,000)

$$0.4(0.6x_1) + 0.7(0.6x_2) + 0.8(0.3x_3) \geq 0.5(600)$$

A Purchasing Problem | Constraints

- The maximum click bound should not be exceeded

$$x_1 \leq 600$$

$$x_2 \leq 300$$

$$x_3 \leq 300$$

$$x_1, x_2, x_3 \geq 0$$

The Complete LP

$$\text{minimize} \quad 65x_1 + 110x_2 + 120x_3$$

$$\text{subject to} \quad 0.35x_1 + 0.3x_2 + 0.7x_3 \geq 500$$

$$0.4(0.35x_1) + 0.7(0.3x_2) + 0.8(0.7x_3) \geq 0.5(500)$$

$$0.6x_1 + 0.6x_2 + 0.3x_3 \geq 600$$

$$0.4(0.6x_1) + 0.7(0.6x_2) + 0.8(0.3x_3) \geq 0.5(600)$$

$$x_1 \leq 600$$

$$x_2 \leq 300$$

$$x_3 \leq 300$$

$$x_1, x_2, x_3 \geq 0$$

Question

Another way to design an efficient advertising campaign is to maximize the number of unique clicks given an advertising budget of \$105,000.

The required number of clicks and unique clicks in each age category must still be met and the maximum number of clicks from each type of ad still applies. How would you change the existing model?

Solution

Our previous objective is going to become a constraint, and the new objective is to **maximize the number of unique clicks**

$$\text{maximize} \quad 0.4x_1 + 0.7x_2 + 0.8x_3$$

$$\text{subject to} \quad 65x_1 + 110x_2 + 120x_3 \leq 105,000$$

$$0.35x_1 + 0.3x_2 + 0.7x_3 \geq 500$$

$$0.4(0.35x_1) + 0.7(0.3x_2) + 0.8(0.7x_3) \geq 0.5(500)$$

$$0.6x_1 + 0.6x_2 + 0.3x_3 \geq 600$$

$$0.4(0.6x_1) + 0.7(0.6x_2) + 0.8(0.3x_3) \geq 0.5(600)$$

$$x_1 \leq 600$$

$$x_2 \leq 300$$

$$x_3 \leq 300$$

$$x_1, x_2, x_3 \geq 0$$

Solving Linear Programs

Graphical Solution Method

Solving Linear Programming Problems

- Graphical method

Only useful for 2 decision variables (maybe 3 if you can handle 3-D chart)

Solving Linear Programming Problems

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- Simplex method

Efficient algorithm to solve LP problems by performing matrix operations

Developed by George Dantzig (1947)

Can be used to solve small LP problems by hand

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Lindo, LPSolve, MS Excel

AMPL/CPLEX/Gurobi: modeling language and "solver" for large and complex LP/IP problems

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- Heuristics (sub-optimal), ...

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- Heuristics (sub-optimal), ...

Solving a Linear Program Graphically

Step 1: Graph the region that satisfies all of the constraints.

Step 2: Draw a contour line of the objective function and determine which direction improves the objective function. A second contour line can be used to determine which direction is improving.

Step 3: Sweep the contour line in the improving direction, generating parallel lines, until it is about to leave the region. The last line intersecting the region has the optimal objective function value. The point(s) where this line intersects the region is optimal.

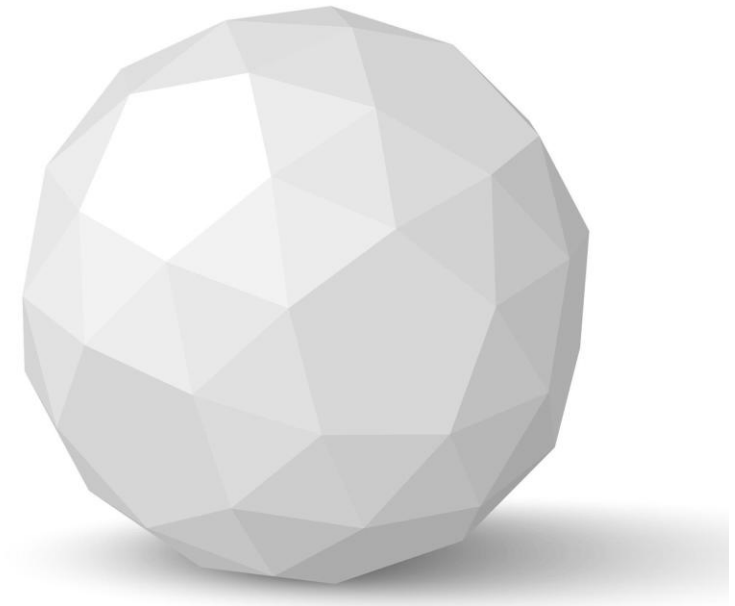
Step 4: Find the coordinates of an optimal point by solving a system of two linear equations from the constraints whose lines pass through this point.

Fun(?) Fact: Feasible Region of an LP

- The feasible region of an LP is always convex and is called a **polyhedron***

*A polyhedron is a three-dimensional geometric figure with flat faces, straight edges, and sharp vertices. It is a solid bounded by flat polygonal faces, where each face is a polygon.

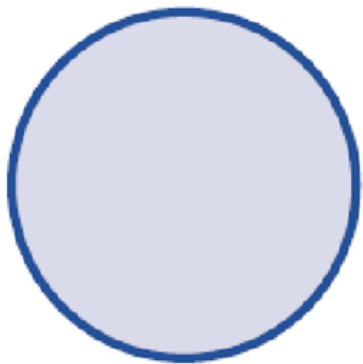
The term "polyhedron" comes from the Greek words "poly" (meaning "many") and "hedra" (meaning "face").



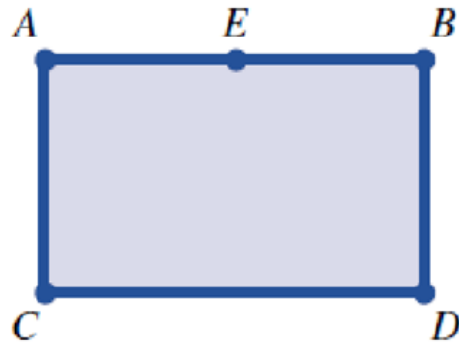
Basic Concept: Convexity

- A set S is convex if, for every two points $x, y \in S$, the segment joining them also belongs to S

S = shaded area



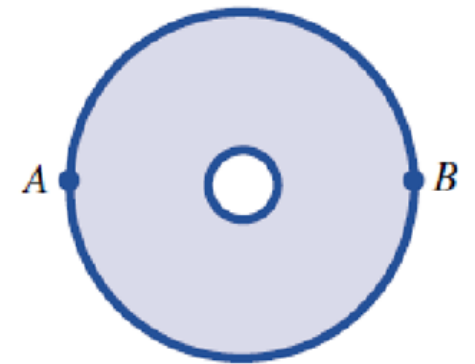
a convex



b convex



c not
convex



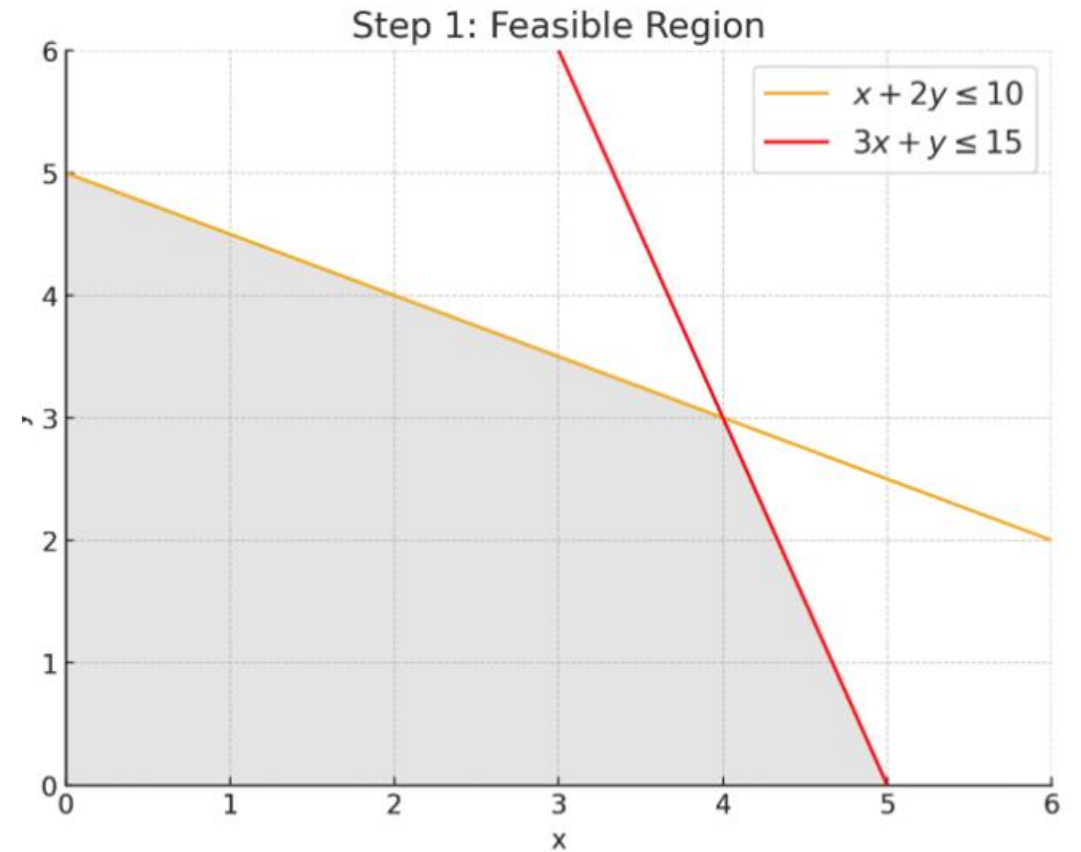
d not
convex

Solving a Linear Program Graphically

Step 1: Graph the region that satisfies all of the constraints.

Example:

- Constraints:
 $x + 2y \leq 10$
 $3x + y \leq 15$
 $x \geq 0$
 $y \geq 0$

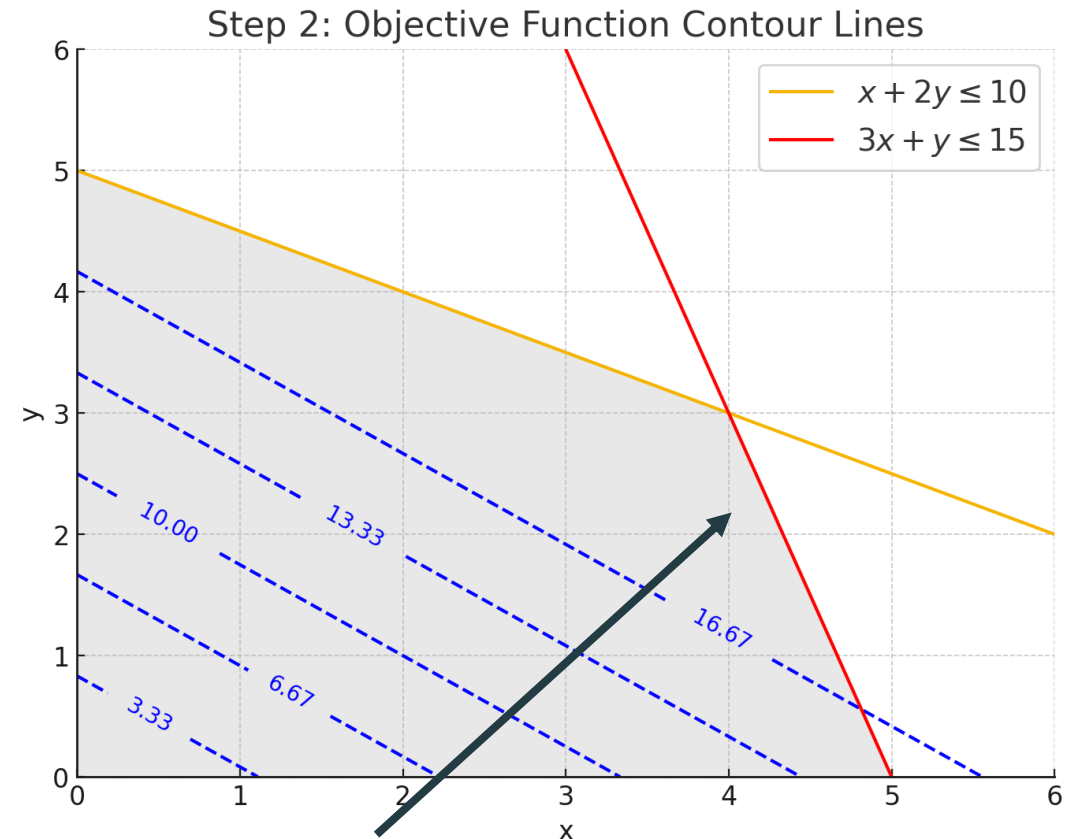


Solving a Linear Program Graphically

Step 2: Draw a contour line of the objective function and determine which direction improves the objective function. A second contour line can be used to determine which direction is improving.

Objective function: $z = 3x + 4y$

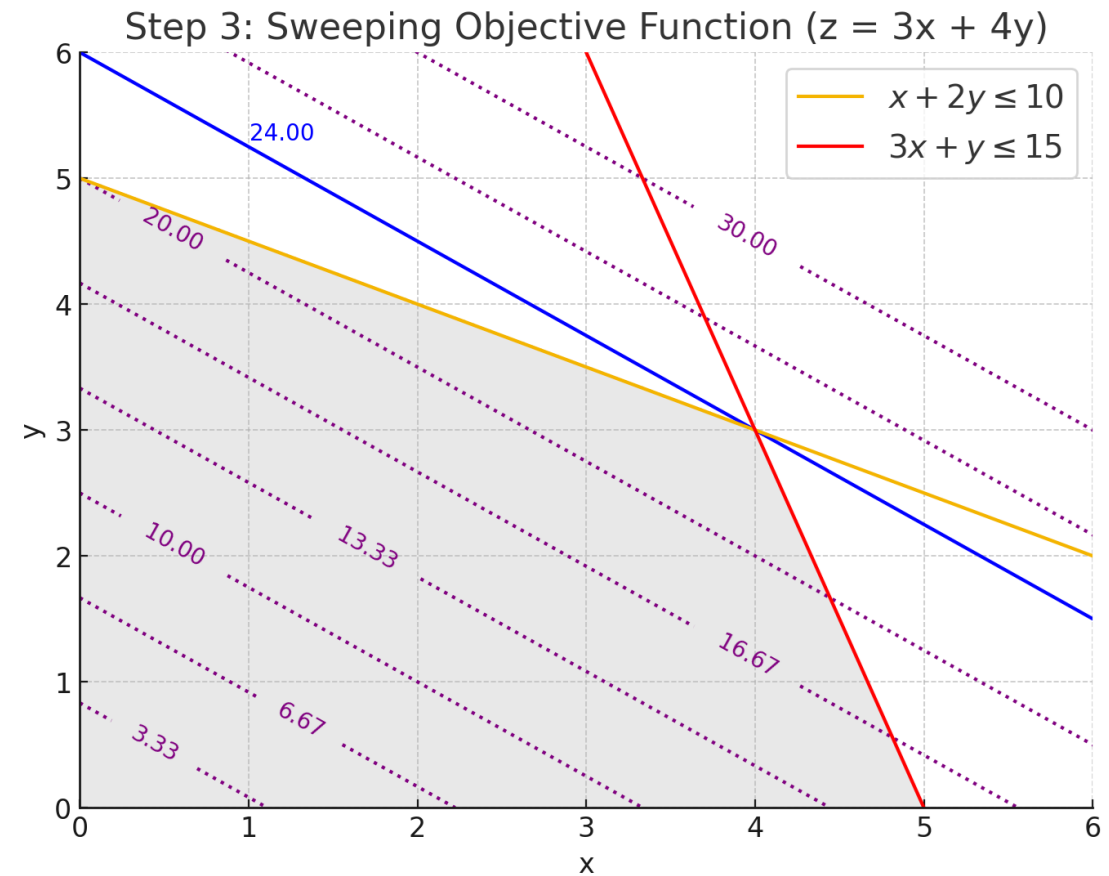
- The improving direction is towards higher values as we aim to maximize z



Solving a Linear Program Graphically

Step 3: Sweep the contour line in the improving direction, generating parallel lines, until it is about to leave the region. The last line intersecting the region has the optimal objective function value. The point(s) where this line intersects the region is optimal.

- Here, solid blue line represents $z=24$, which is the maximum value achievable within the feasible region.



Solving a Linear Program Graphically

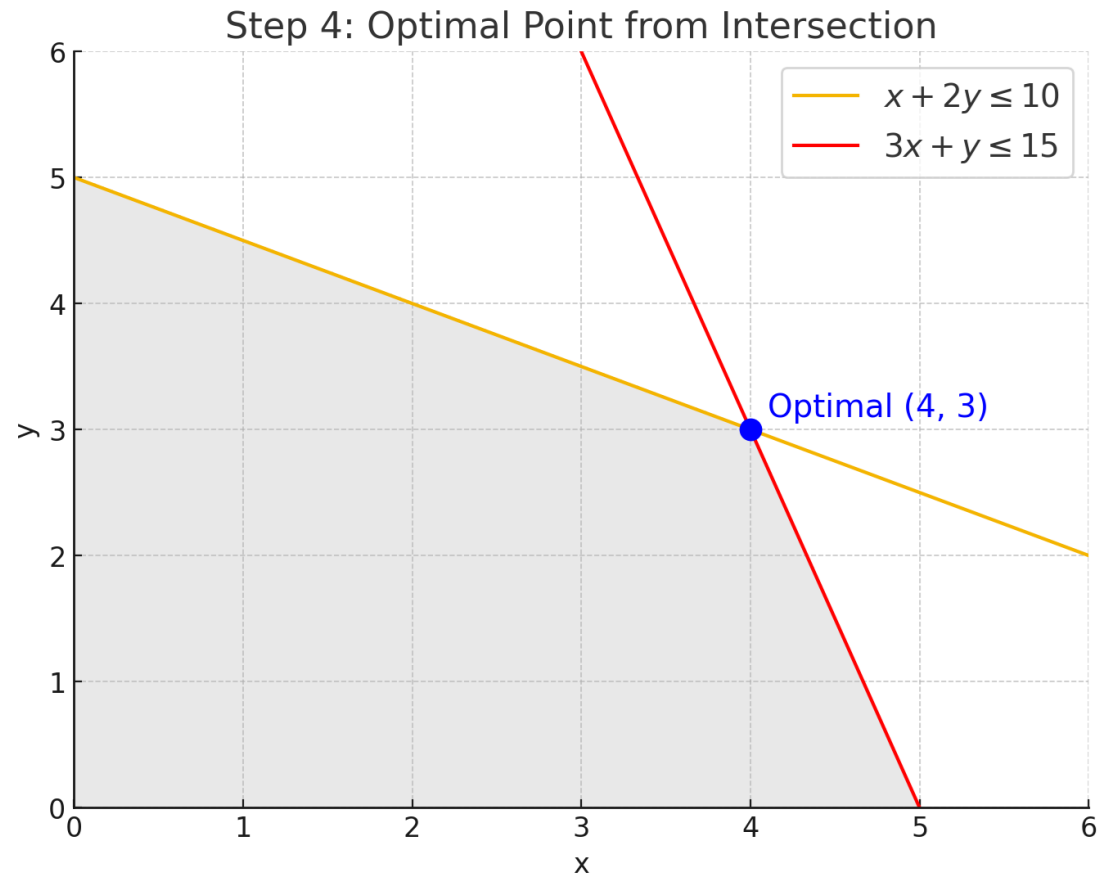
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- Solve the system:

$$x + 2y = 10$$

$$3x + y = 15$$

$$\text{Solution} \Rightarrow (x, y) = (4, 3)$$



Graphical Solution Method

Let's look at this LP:

$$\text{Maximize}_{x_1, x_2} 3x_1 + 2x_2$$

s. t.

$$x_1 + x_2 \leq 80$$

$$2x_1 + x_2 \leq 100$$

$$x_1 \leq 40$$

$$x_1, x_2 \geq 0$$

Graphical Solution Method

Try and draw it:

$$\text{Maximize}_{x_1, x_2} 3x_1 + 2x_2$$

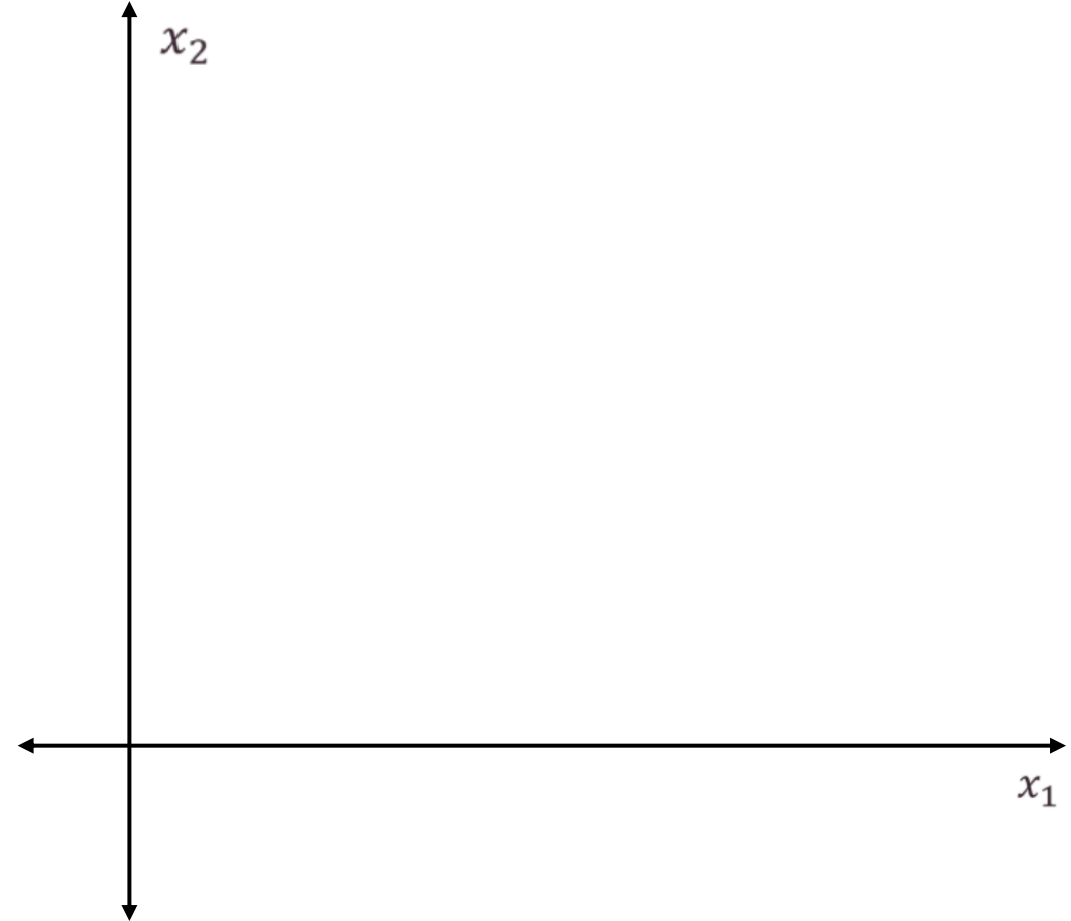
s. t.

$$x_1 + x_2 \leq 80$$

$$2x_1 + x_2 \leq 100$$

$$x_1 \leq 40$$

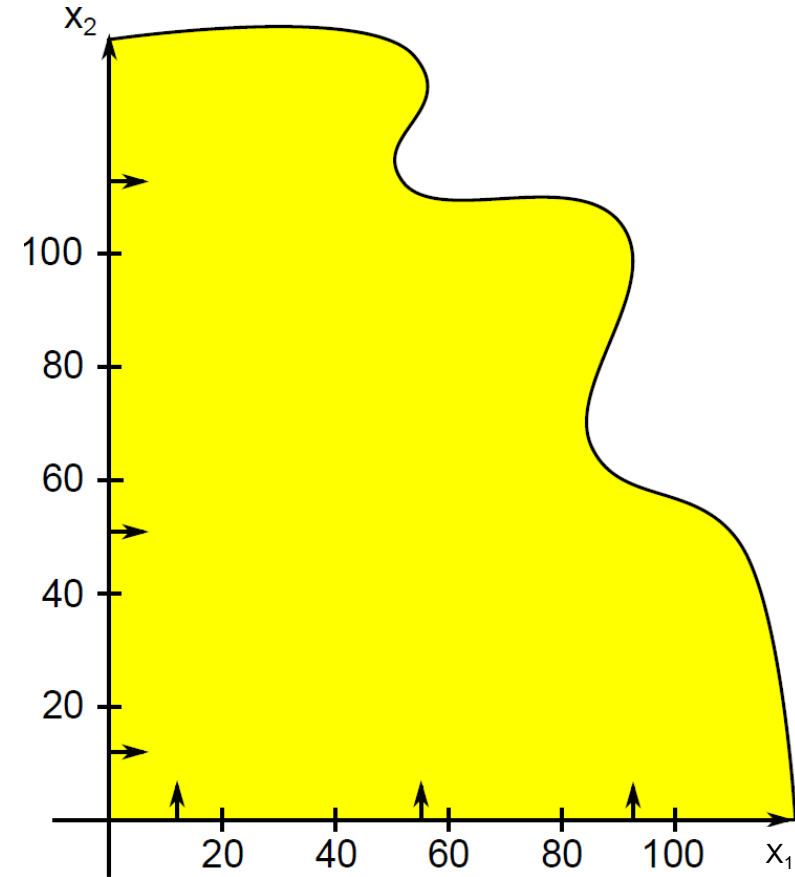
$$x_1, x_2 \geq 0$$



Graphical Solution Method

$$\begin{aligned} & \text{Maximize}_{x_1, x_2} 3x_1 + 2x_2 \\ & \text{s.t.} \end{aligned}$$

$$x_1, x_2 \geq 0$$



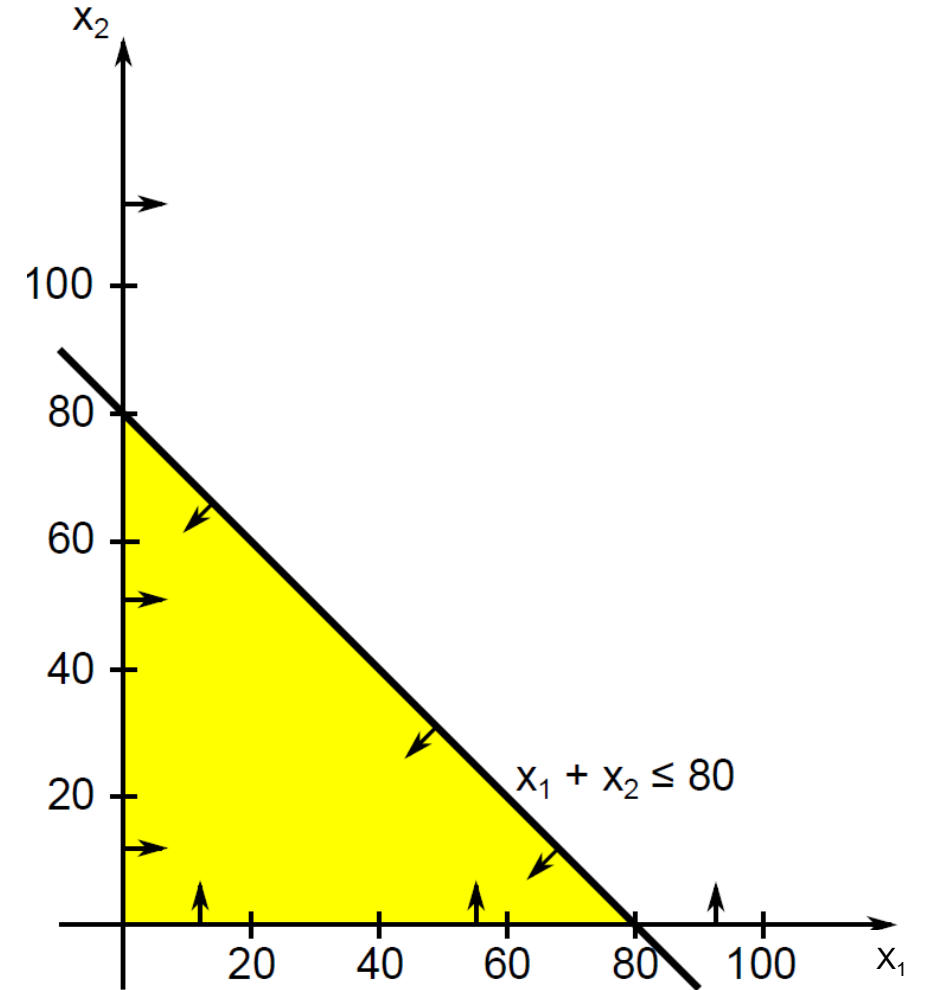
Graphical Solution Method

Maximize _{x_1, x_2} $3x_1 + 2x_2$

s. t.

$$x_1 + x_2 \leq 80$$

$$x_1, x_2 \geq 0$$



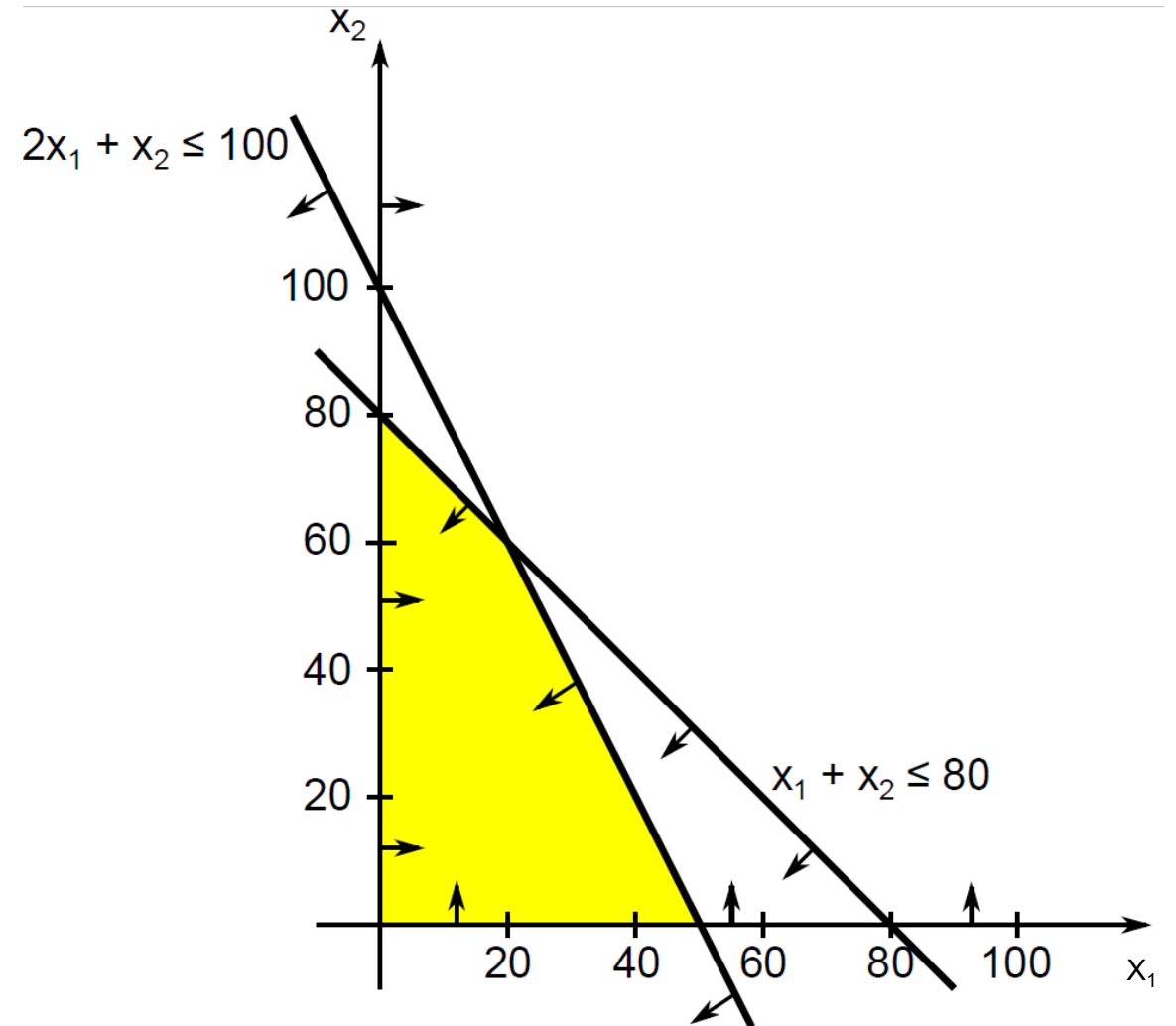
Graphical Solution Method

Maximize _{x_1, x_2} $3x_1 + 2x_2$
s. t.

$$x_1 + x_2 \leq 80$$

$$2x_1 + x_2 \leq 100$$

$$x_1, x_2 \geq 0$$



Graphical Solution Method

Let's look at this LP:

$$\text{Maximize}_{x_1, x_2} 3x_1 + 2x_2$$

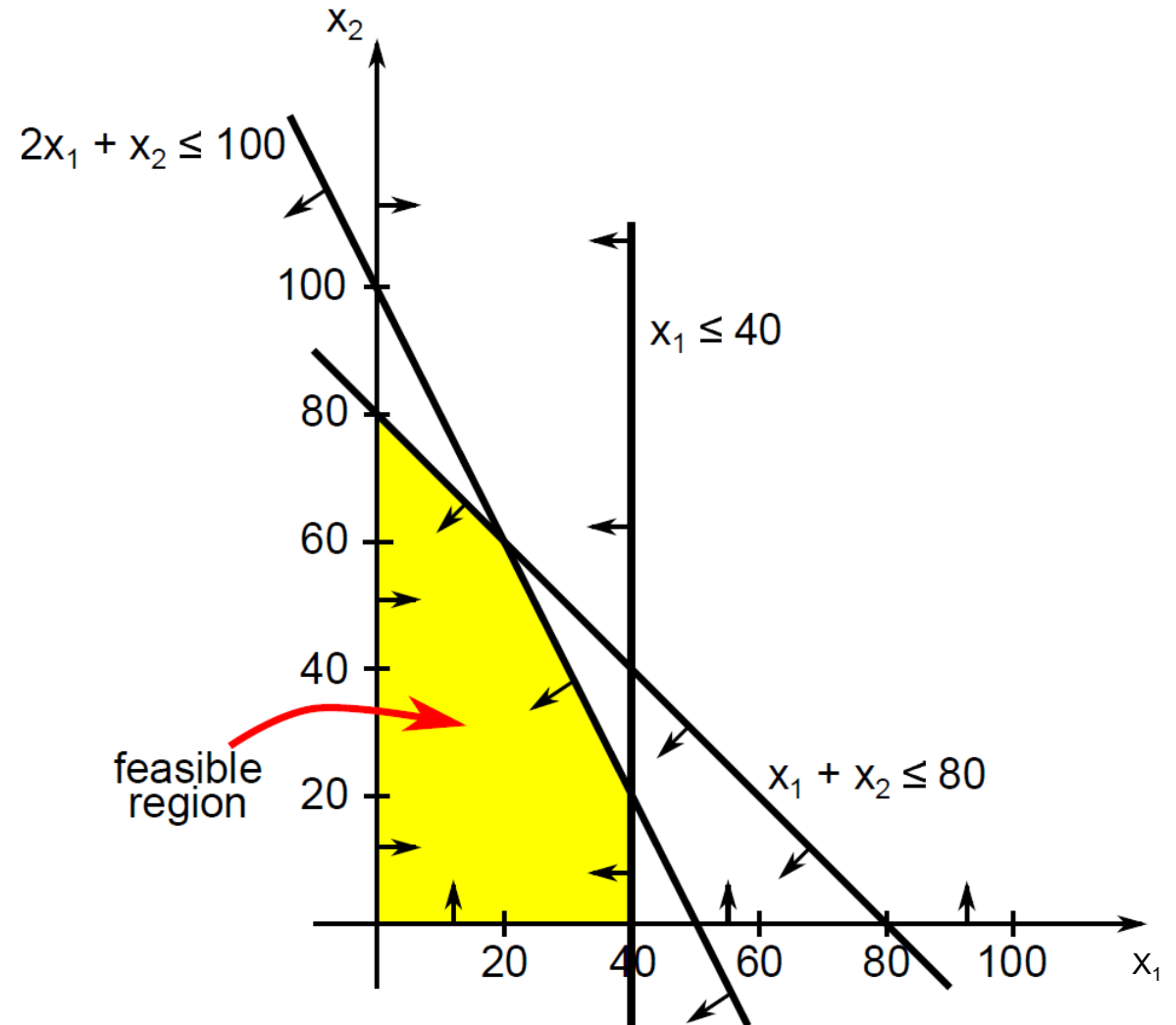
s. t.

$$x_1 + x_2 \leq 80$$

$$2x_1 + x_2 \leq 100$$

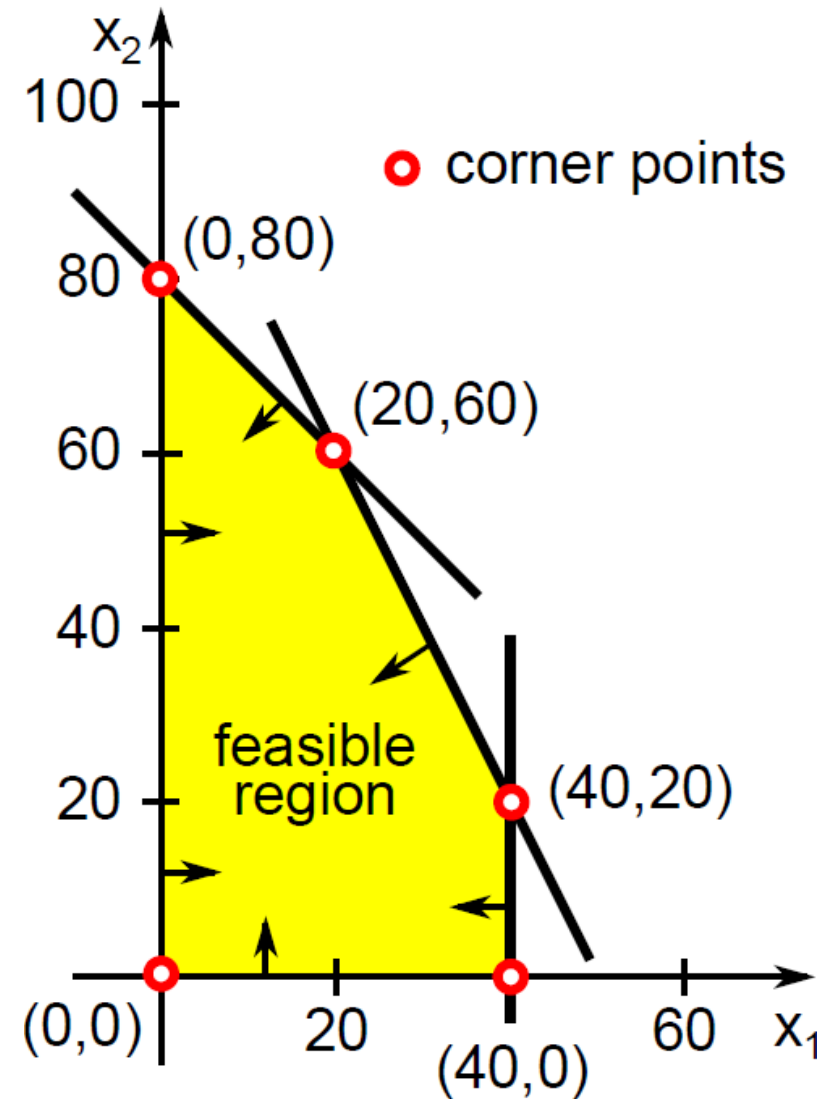
$$x_1 \leq 40$$

$$x_1, x_2 \geq 0$$



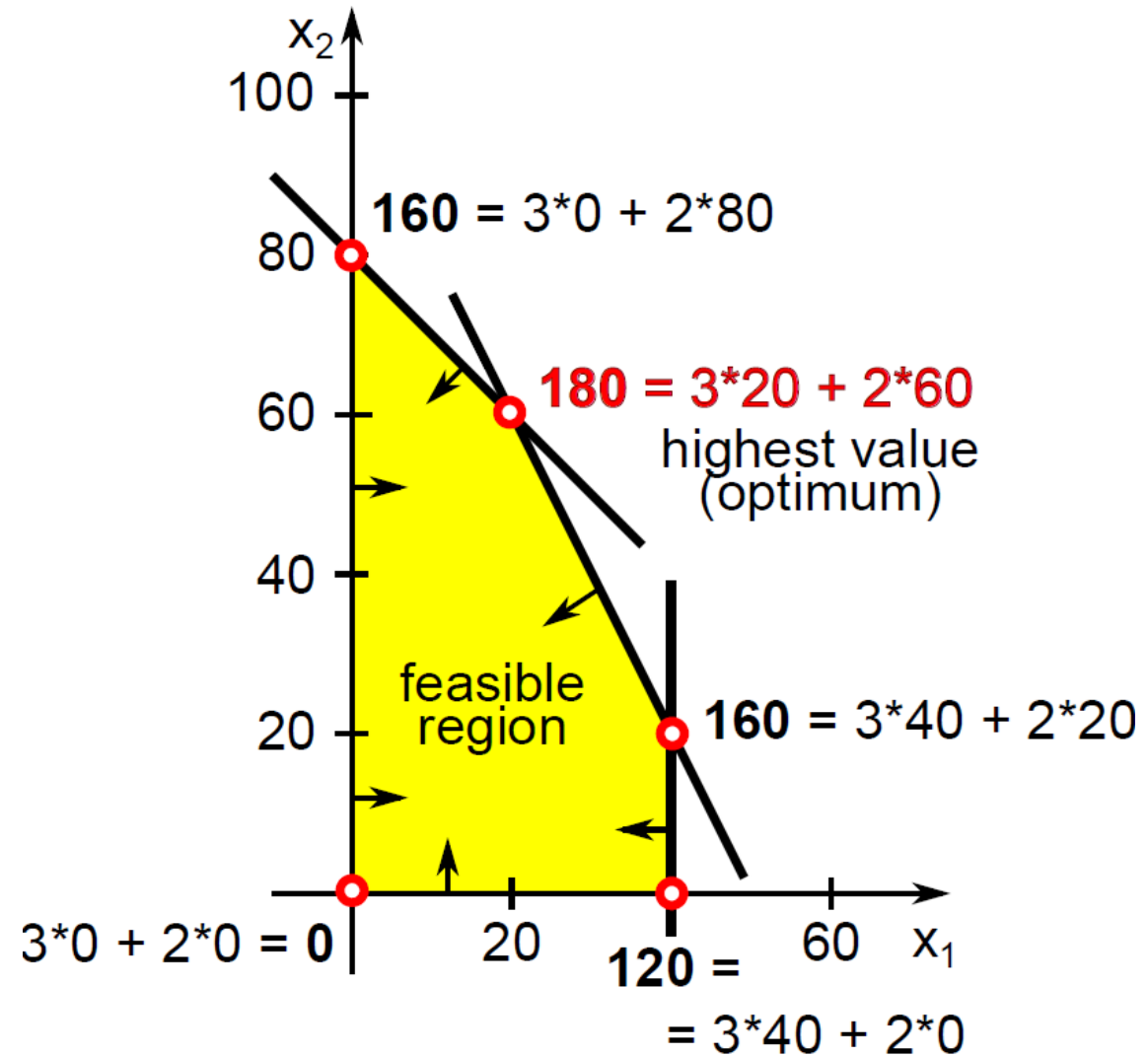
Fun? Fact

If an LP has an optimal solution, it lies at one of the corner point of the feasible region!



Optimal Solution

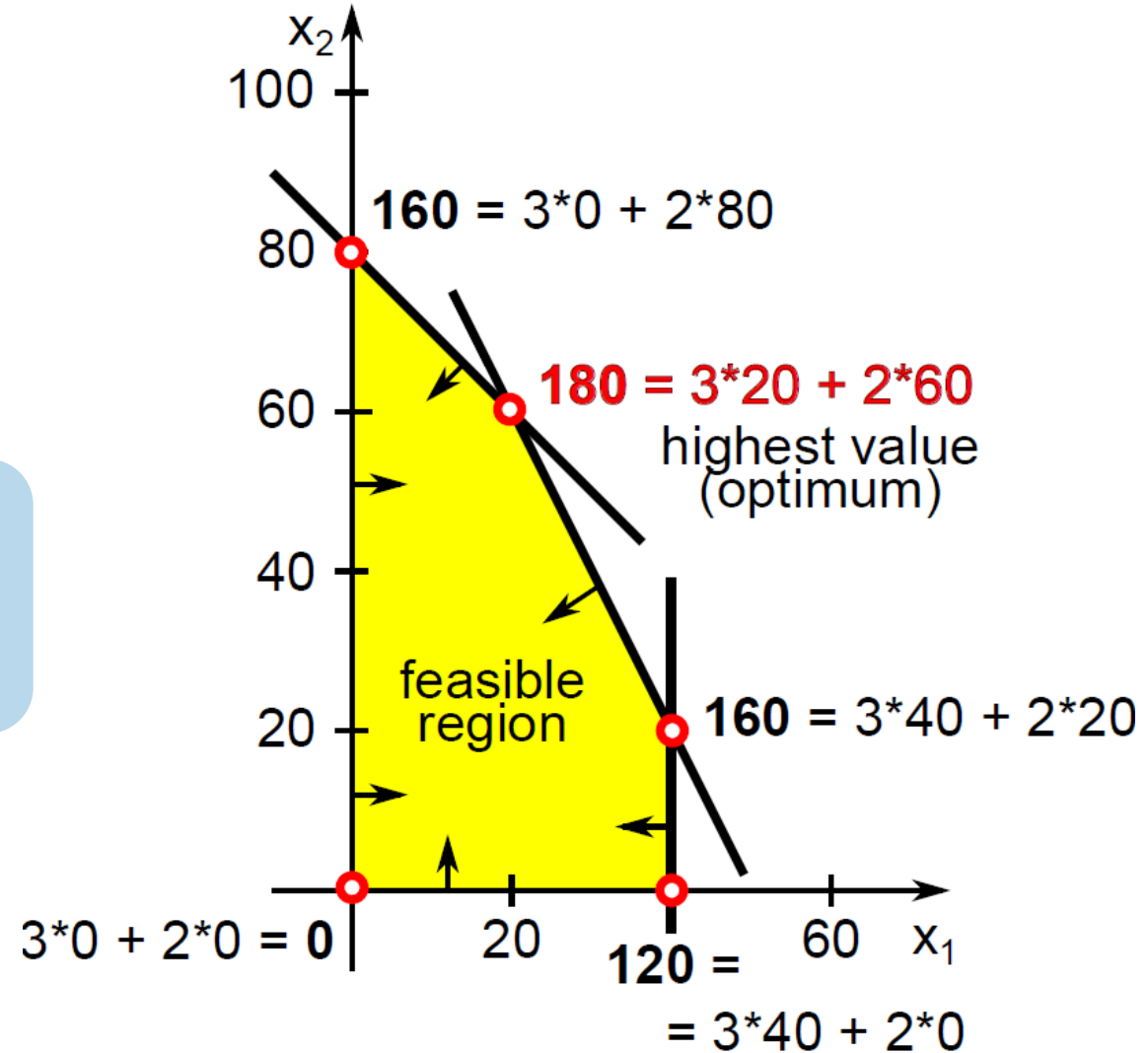
One can check all the corner points...



Optimal Solution

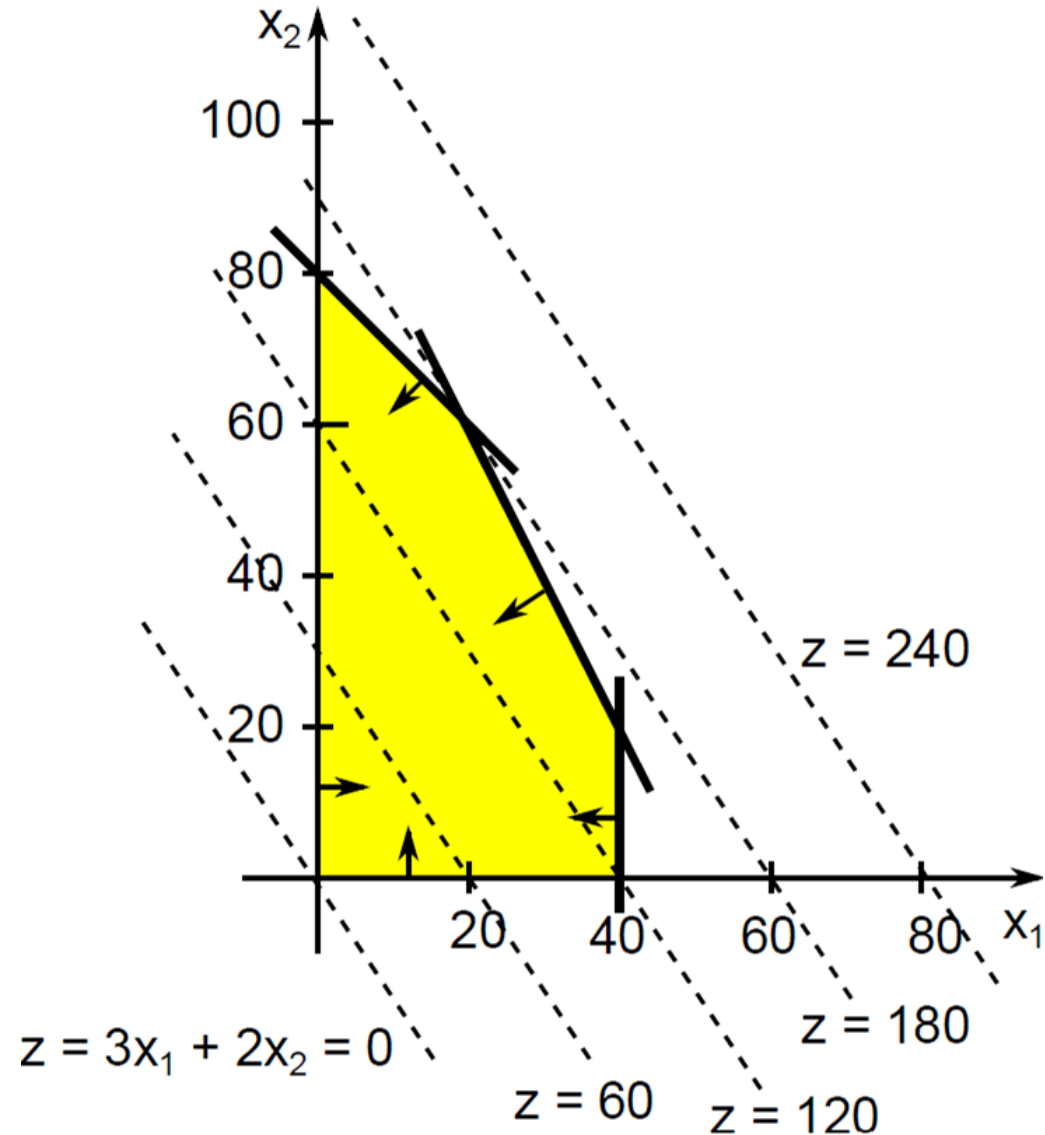
One can check all the corner points...

There may be too many corner points!



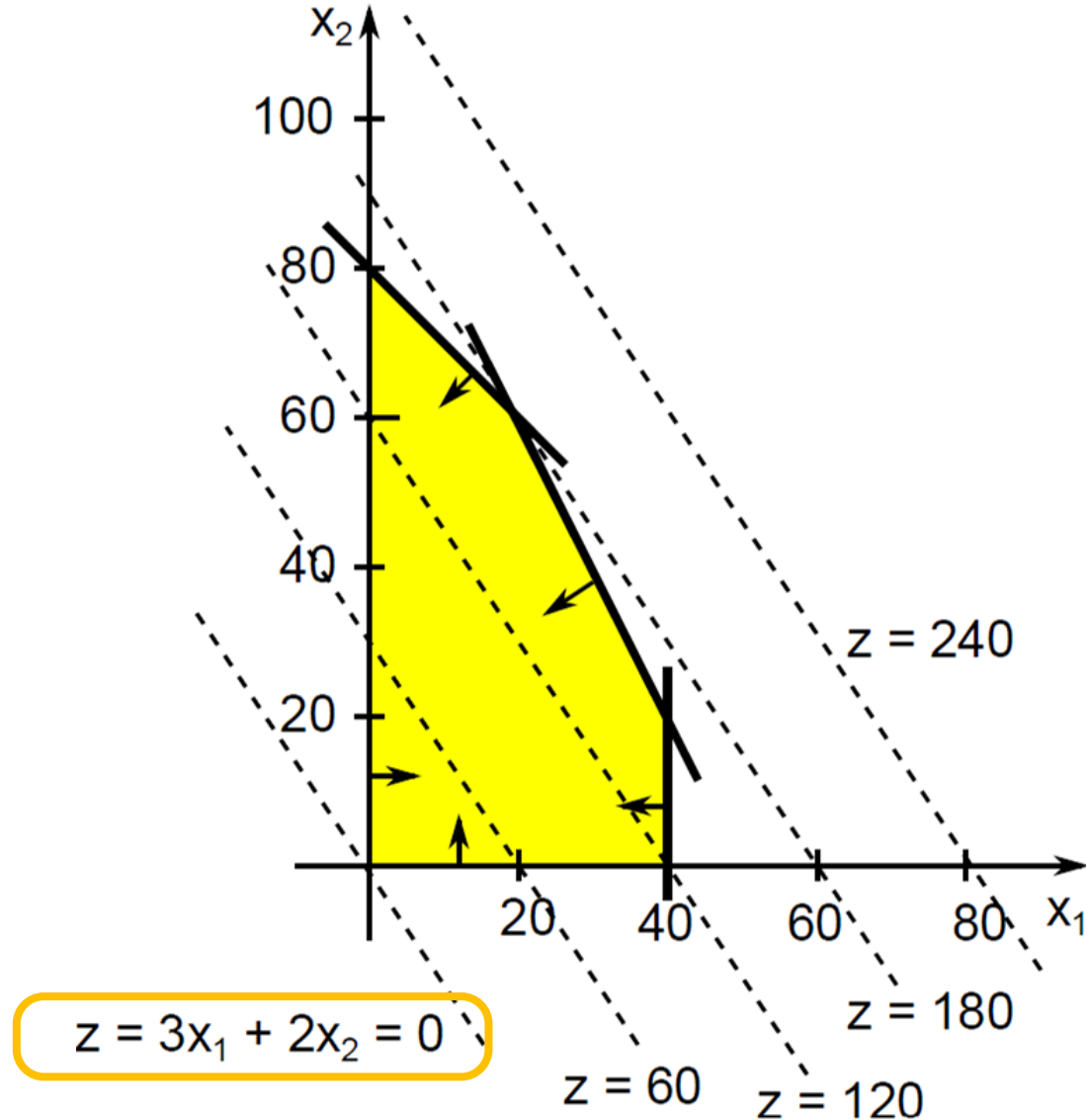
Iso-value Line

We can use the iso-cost
(iso-value) line to find the
optimal solution



Iso-value Line

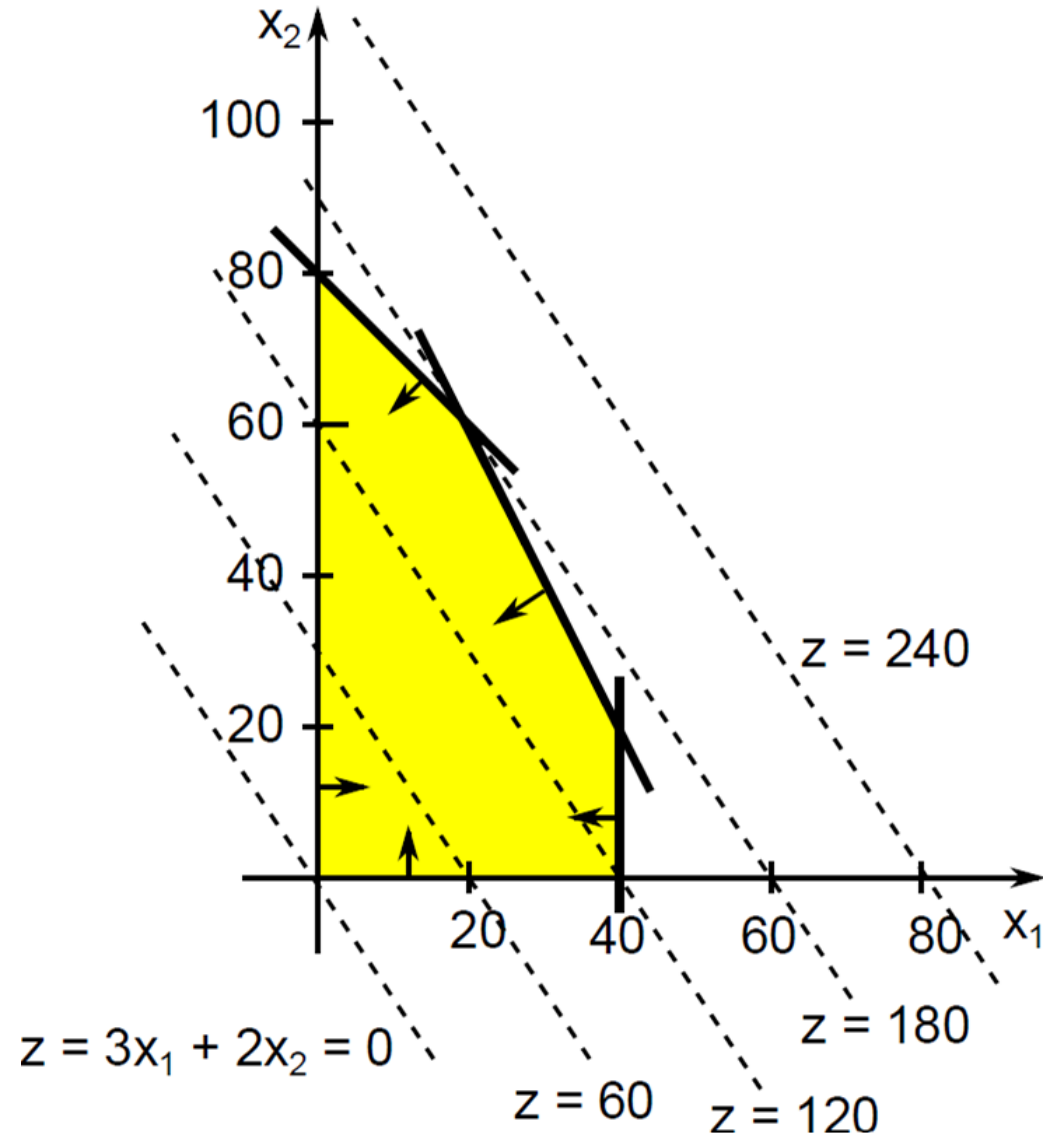
We can use the iso-cost
(iso-value) line to find the
optimal solution



Iso-value Line

We can use the iso-cost
(iso-value) line to find the
optimal solution

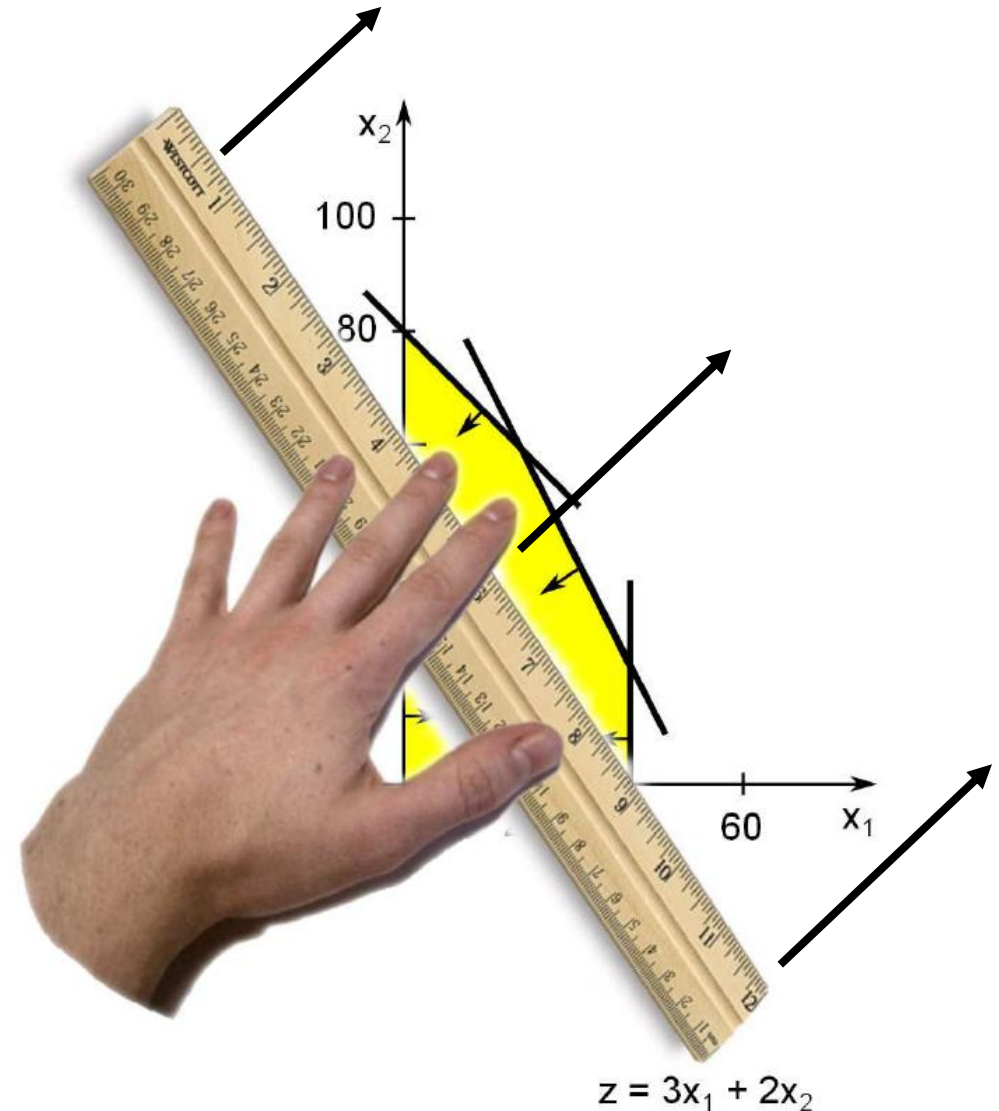
Slide the line in the
increasing direction for
maximization!



Graphical Solution

We can use the iso-cost (iso-value) line to find the optimal solution

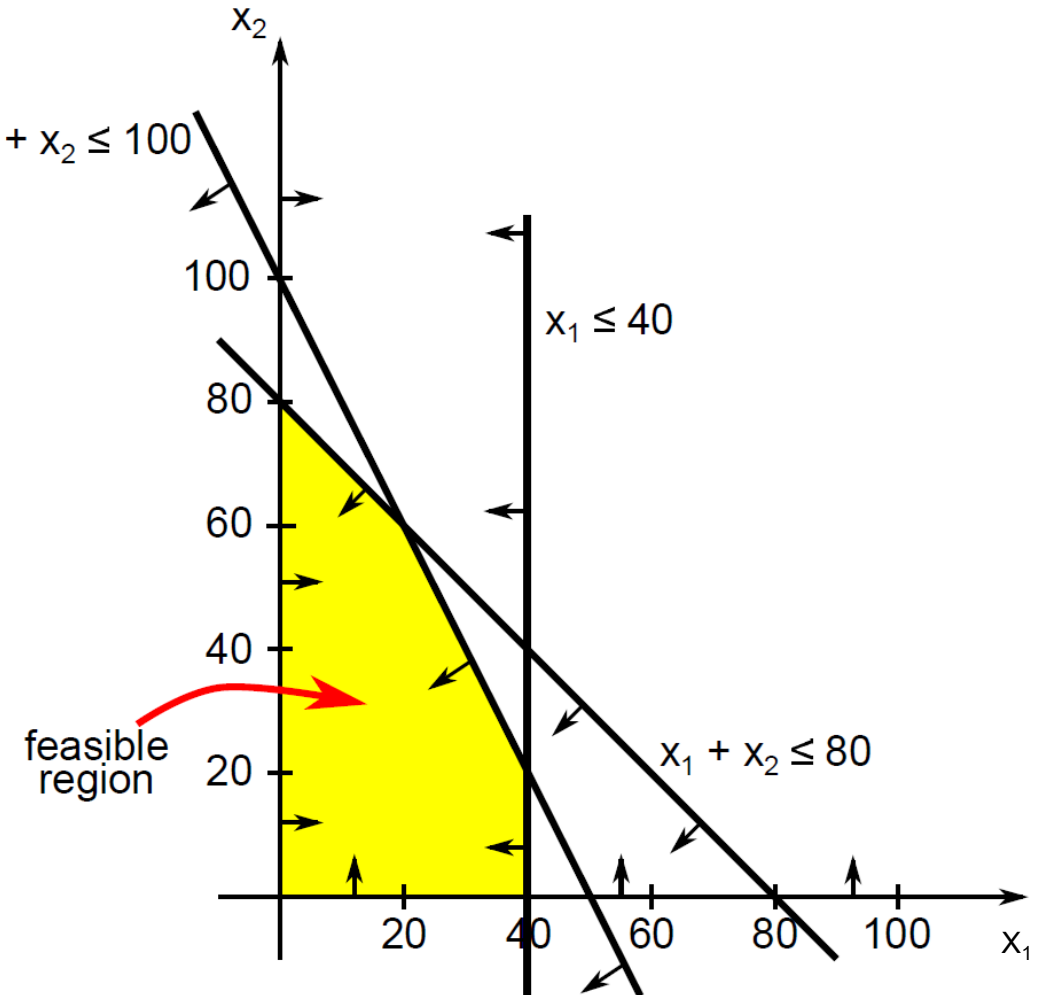
Slide the line in the increasing direction for maximization!



Multiple Optimal Solutions

Suppose the feasibility set is:

$$\begin{cases} x_1 + x_2 \leq 80, & 2x_1 + x_2 \leq 100, \\ x_1 \leq 40, & x_1, x_2 \geq 0 \end{cases}$$



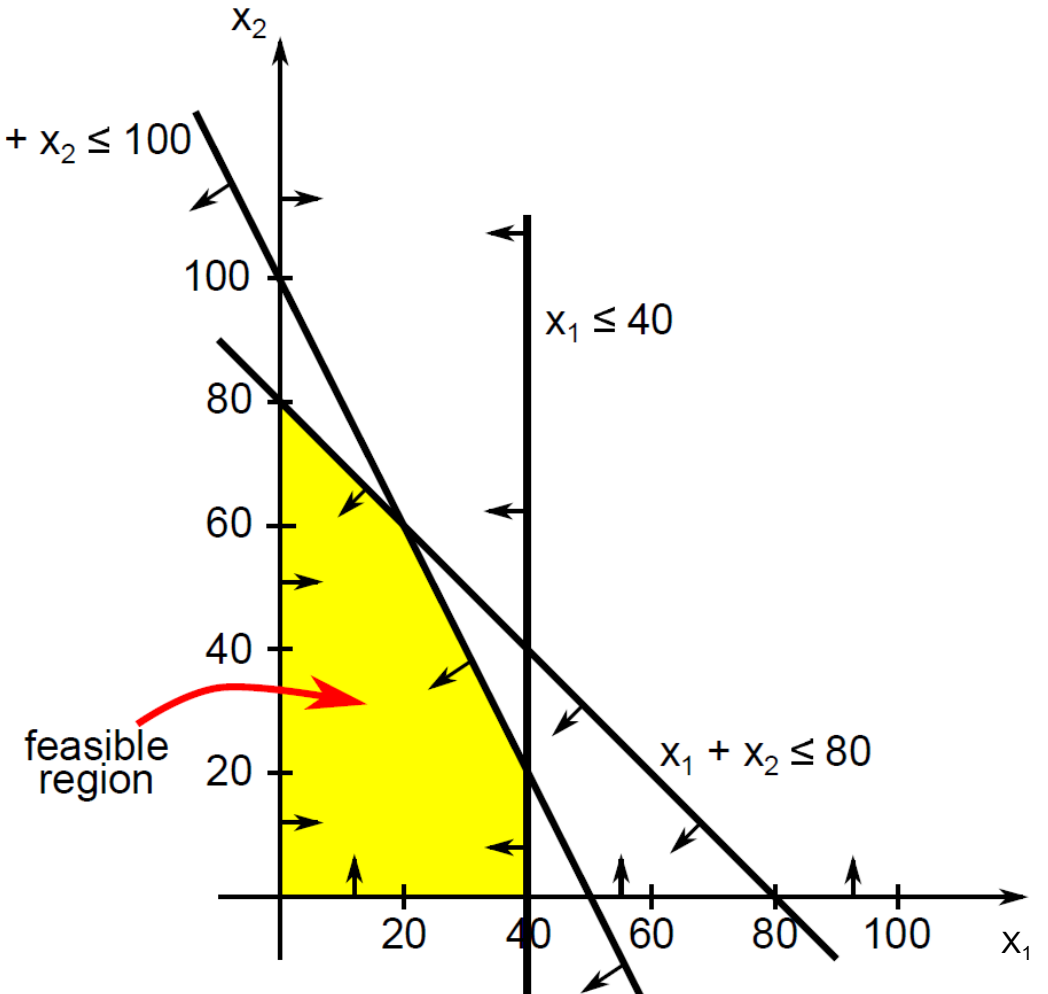
Multiple Optimal Solutions

Suppose the feasibility set is:

$$\{x_1 + x_2 \leq 80, \quad 2x_1 + x_2 \leq 100, \quad x_1 \leq 40, \quad x_1, x_2 \geq 0\}$$

Consider objective function:
 $\max x_1 + x_2$

and try using the iso-value line!



Multiple Optimal Solutions

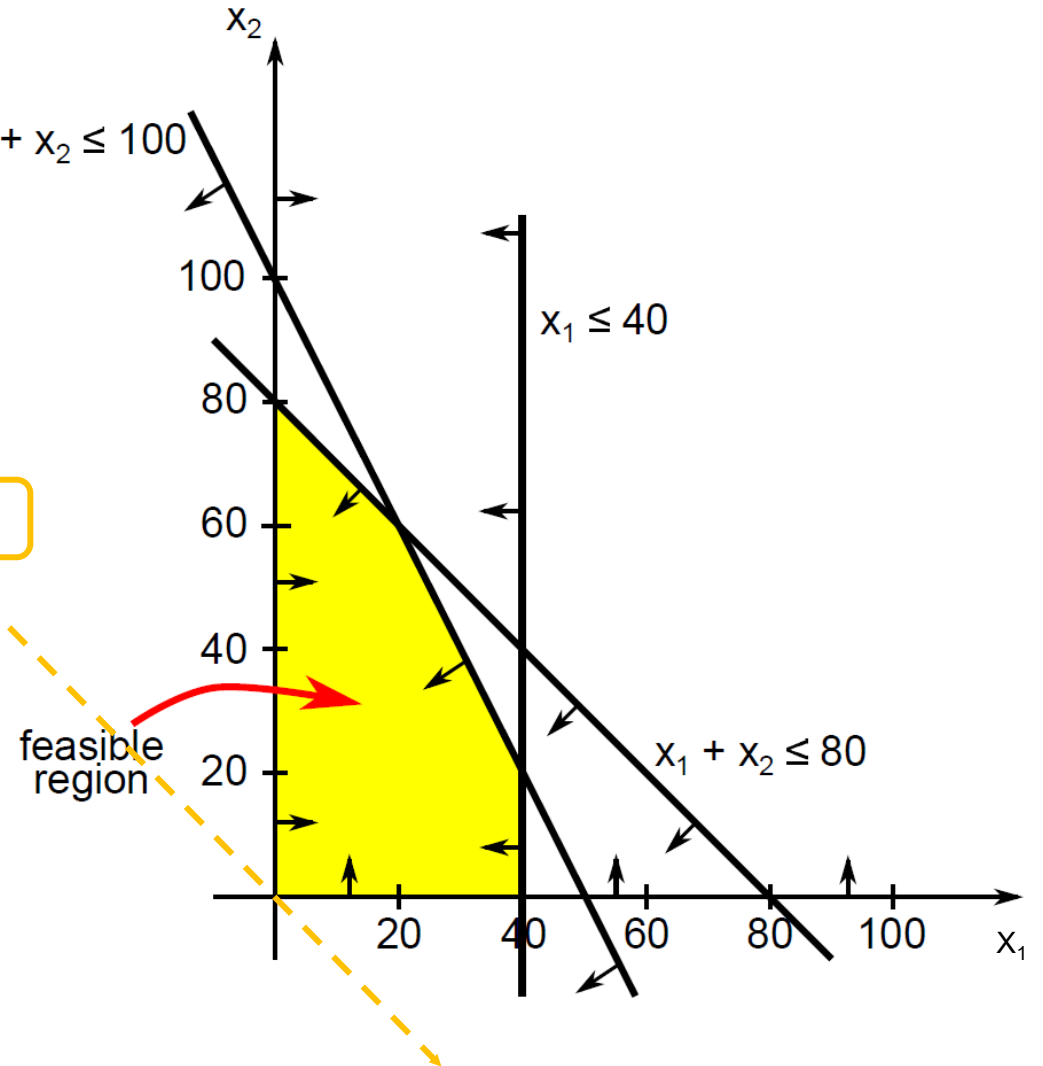
Suppose the feasibility set is:

$$\{x_1 + x_2 \leq 80, \quad 2x_1 + x_2 \leq 100, \quad x_1 \leq 40, \quad x_1, x_2 \geq 0\}$$

Consider objective function:
 $\max x_1 + x_2$

and try using the iso-value line!

$$x_1 + x_2 = 0$$



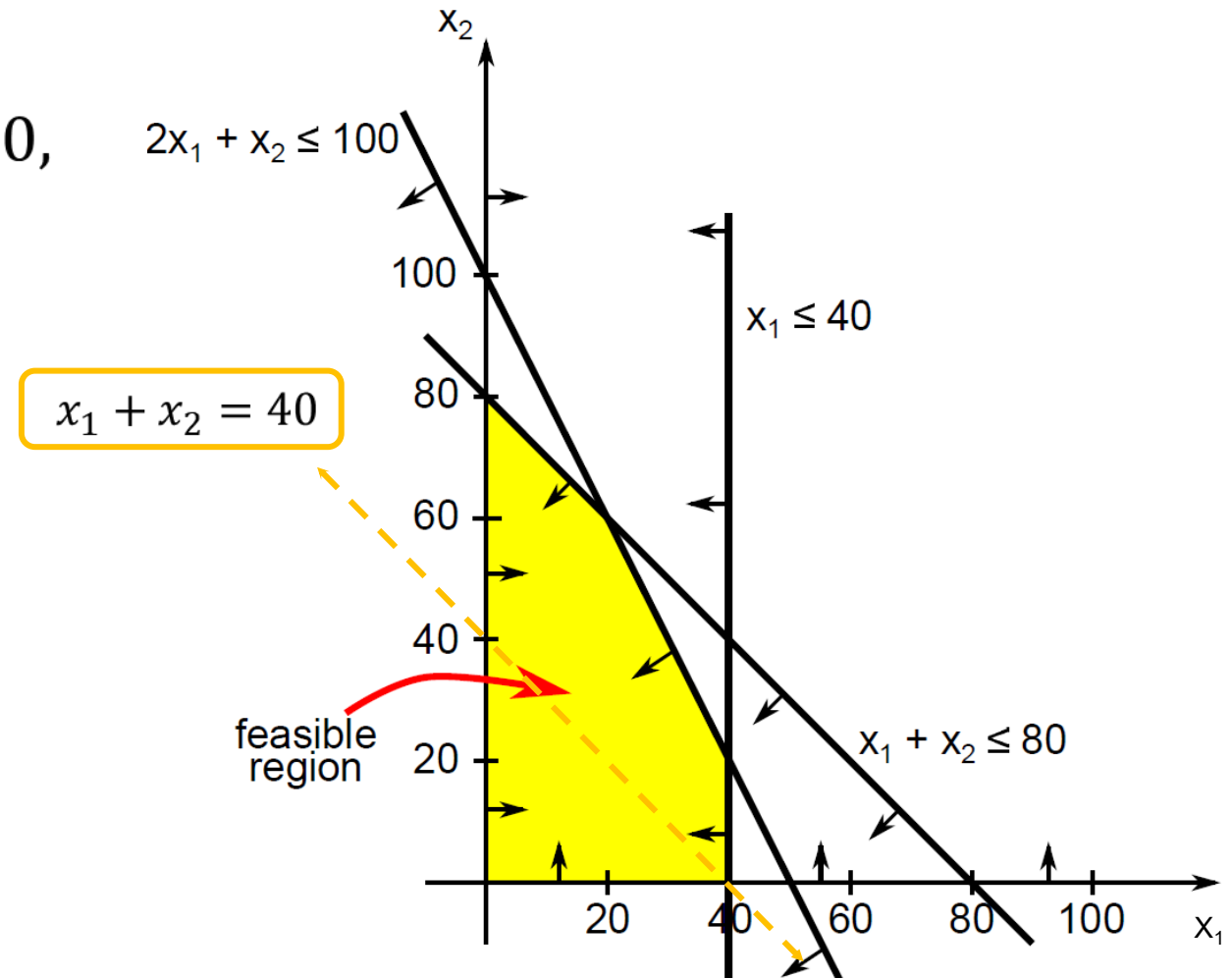
Multiple Optimal Solutions

Suppose the feasibility set is:

$$\{x_1 + x_2 \leq 80, \quad 2x_1 + x_2 \leq 100, \quad x_1 \leq 40, \quad x_1, x_2 \geq 0\}$$

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 $\max x_1 + x_2$

and try using the iso-value line!



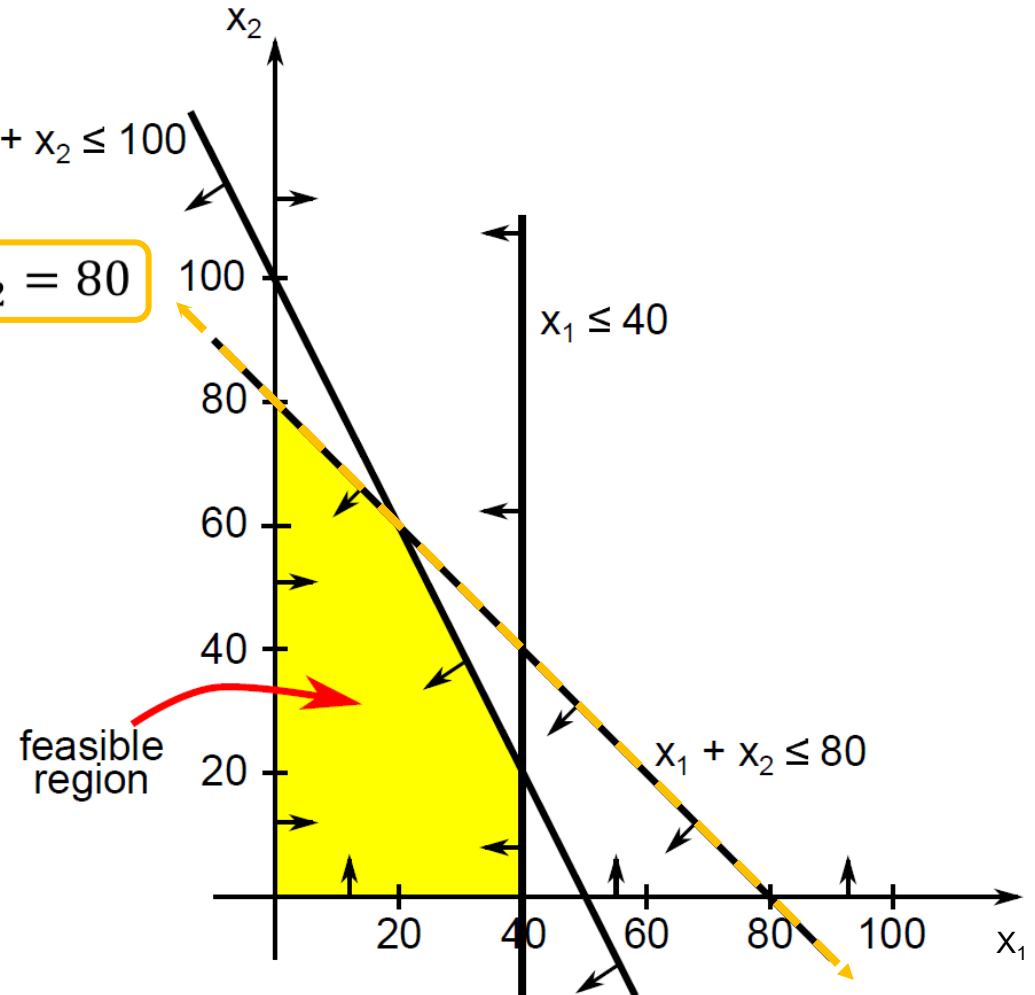
Multiple Optimal Solutions

Suppose the feasibility set is:

$$\{x_1 + x_2 \leq 80, \quad 2x_1 + x_2 \leq 100, \quad x_1 \leq 40, \quad x_1, x_2 \geq 0\}$$

Consider objective function:
 $\max x_1 + x_2$

and try using the iso-value line!



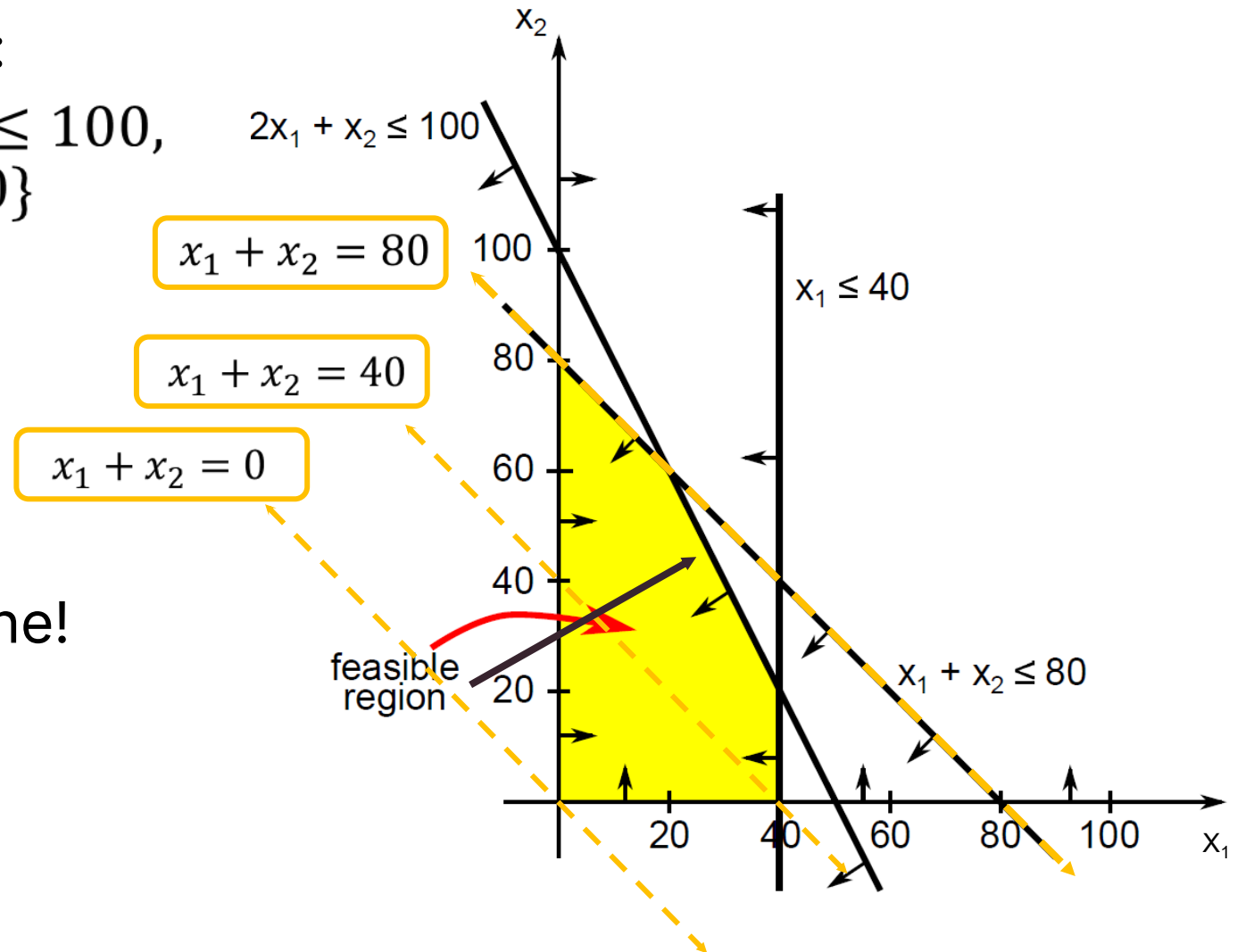
Multiple Optimal Solutions

Suppose the feasibility set is:

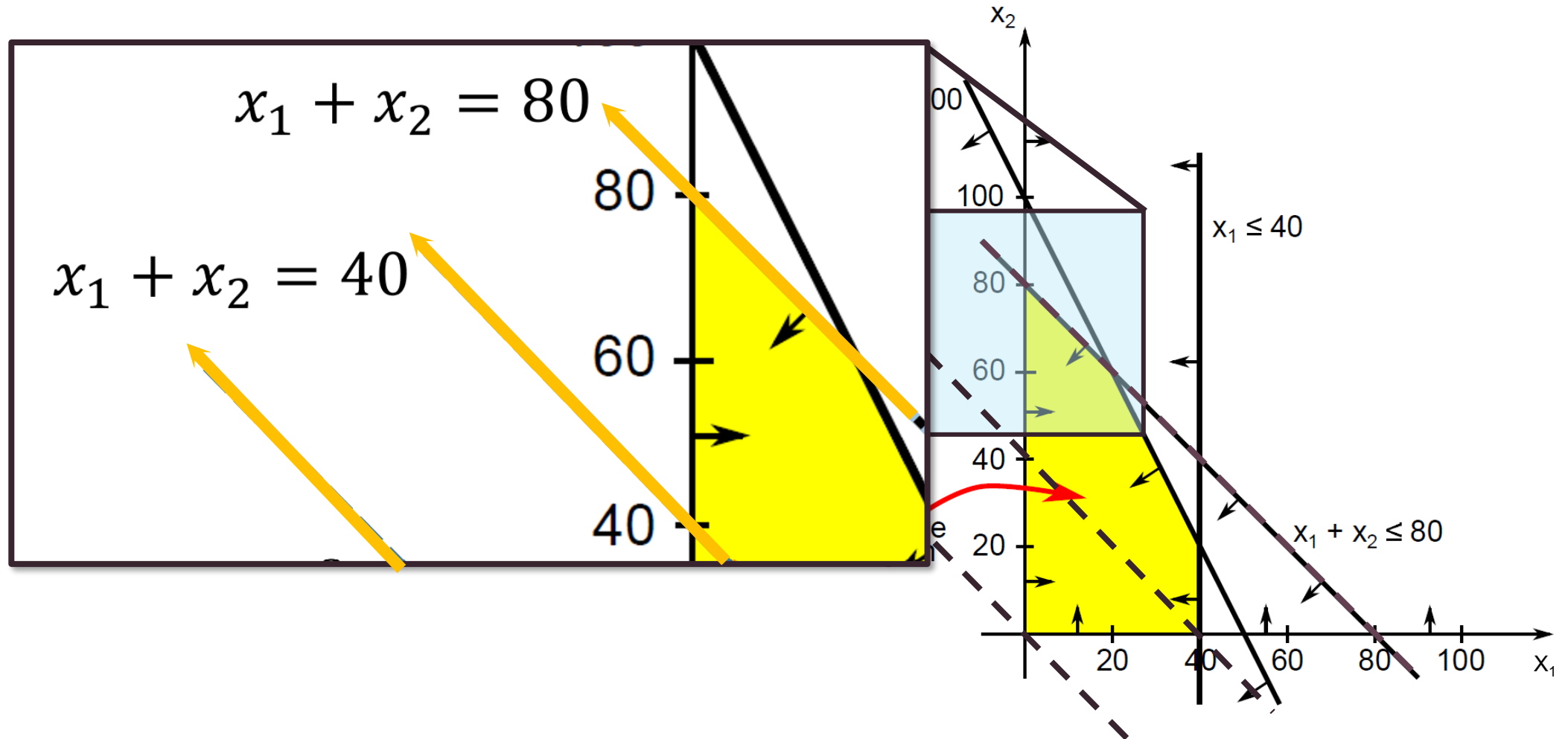
$$\{x_1 + x_2 \leq 80, \quad 2x_1 + x_2 \leq 100, \quad x_1 \leq 40, \quad x_1, x_2 \geq 0\}$$

Consider objective function:
 $\max x_1 + x_2$

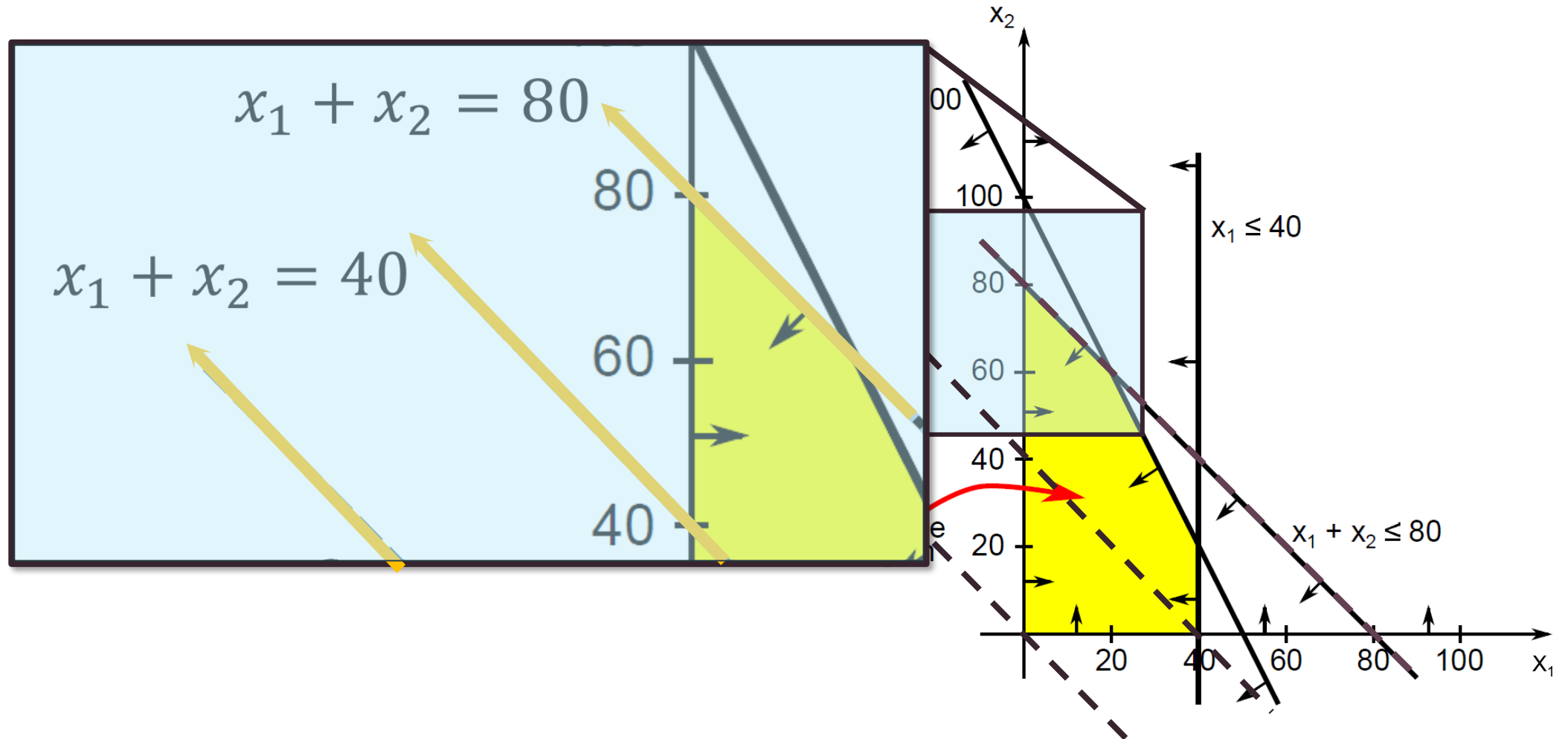
and try using the iso-value line!



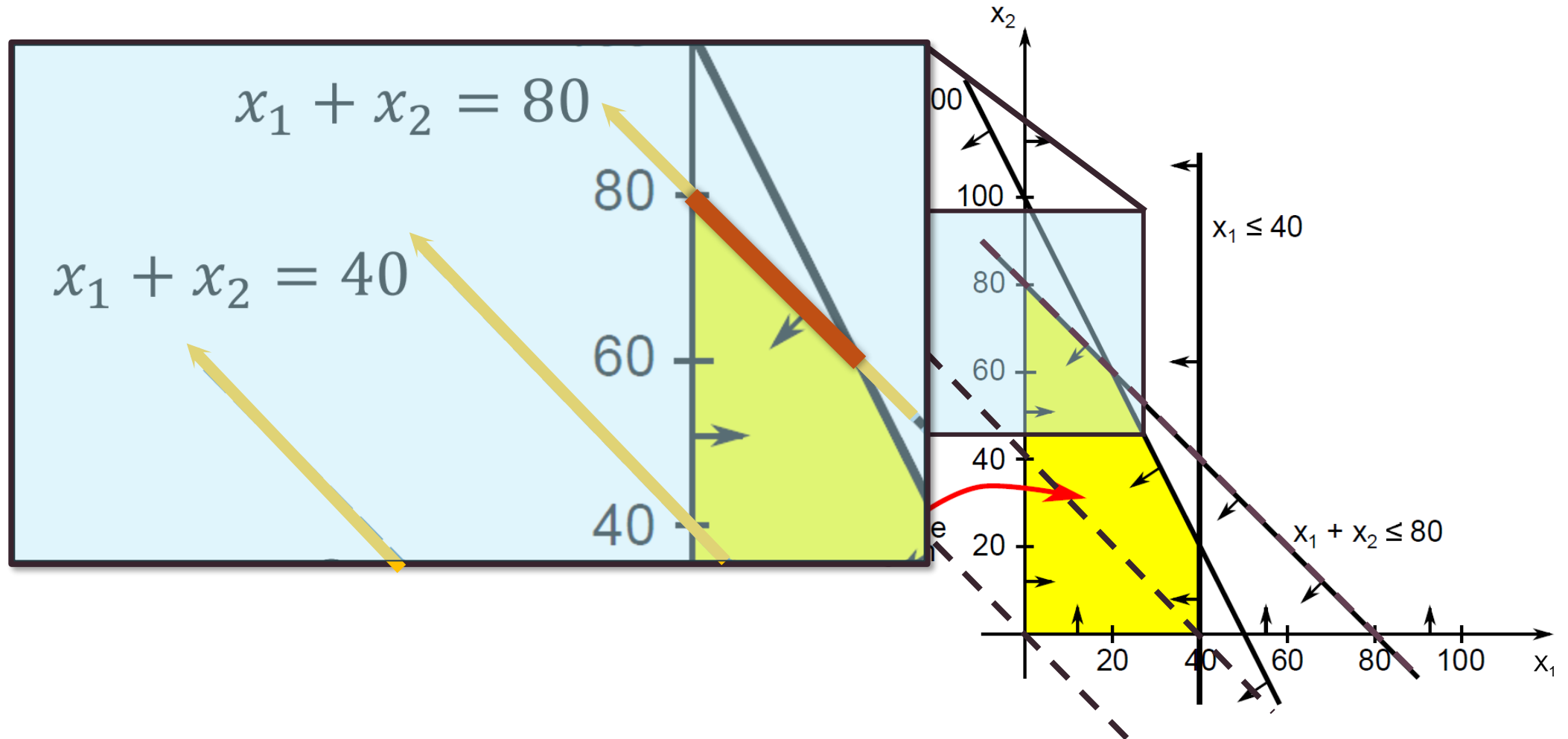
Multiple Optimal Solutions



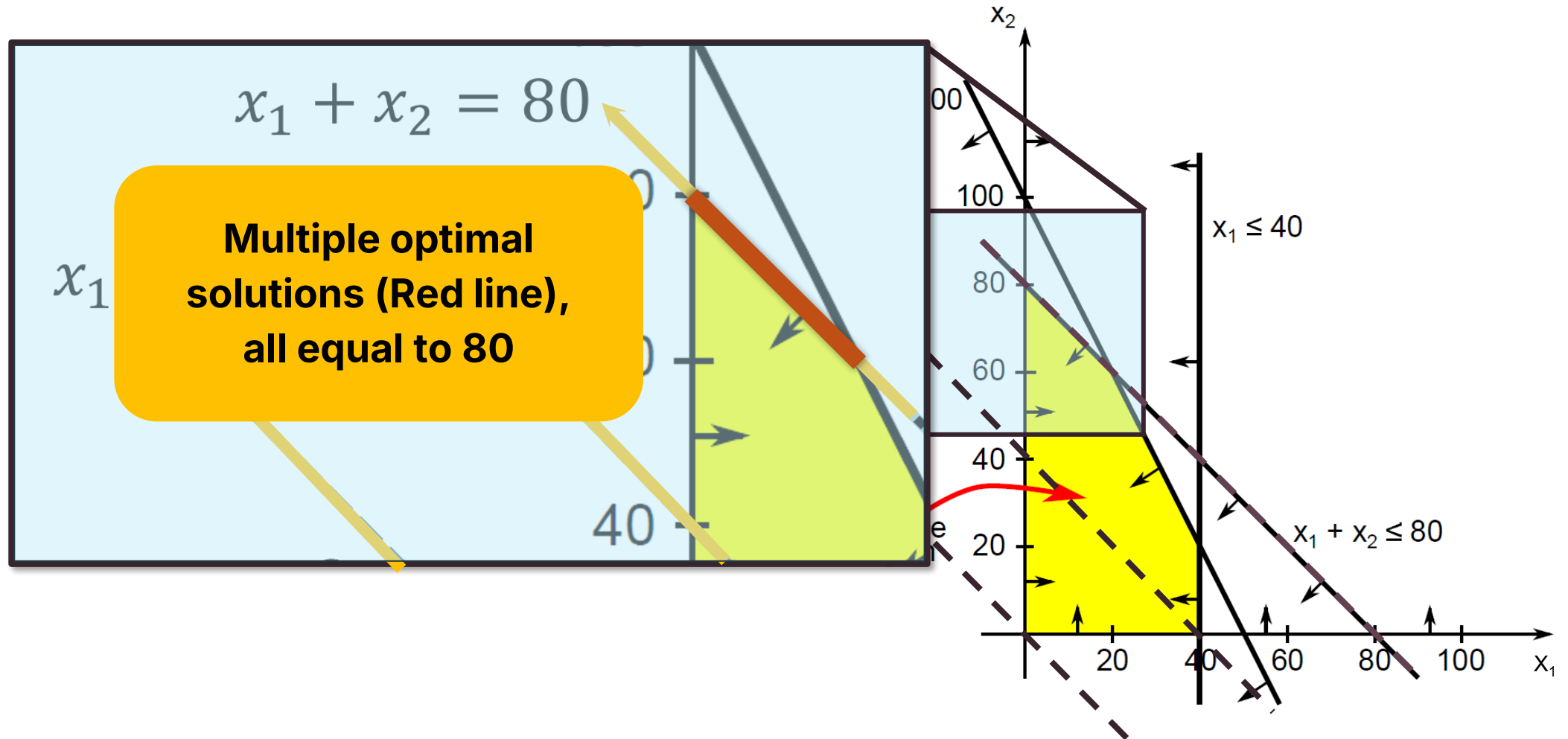
Multiple Optimal Solutions



Multiple Optimal Solutions



Multiple Optimal Solutions



Example | Graphical Solution Method

Let's try to solve this LP:

$$\min z = 50x_1 + 100x_2$$

$$\text{s.t.} \quad 7x_1 + 2x_2 \geq 28$$

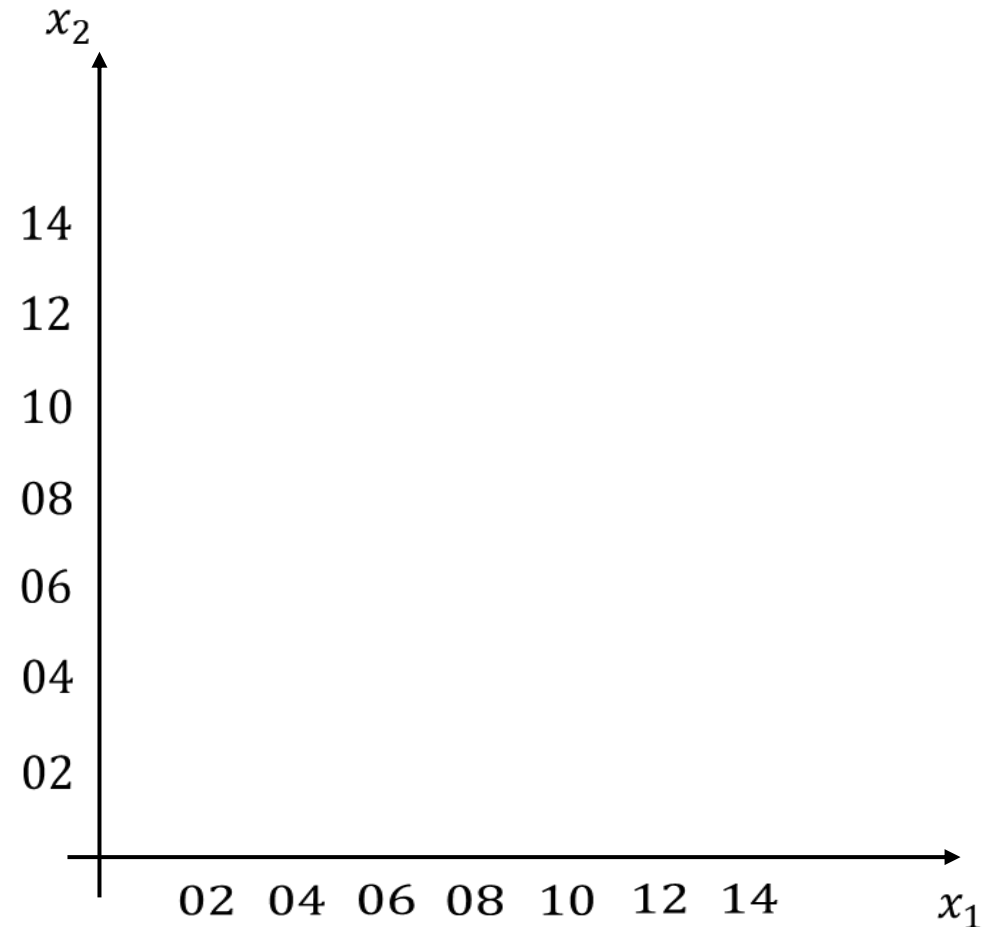
$$2x_1 + 12x_2 \geq 24$$

$$x_1, x_2 \geq 0$$

Example | Graphical Solution Method

Let's try to solve this LP:

$$\begin{array}{ll}\min z = & 50x_1 + 100x_2 \\ \text{s.t.} & 7x_1 + 2x_2 \geq 28 \\ & 2x_1 + 12x_2 \geq 24 \\ & x_1, x_2 \geq 0\end{array}$$

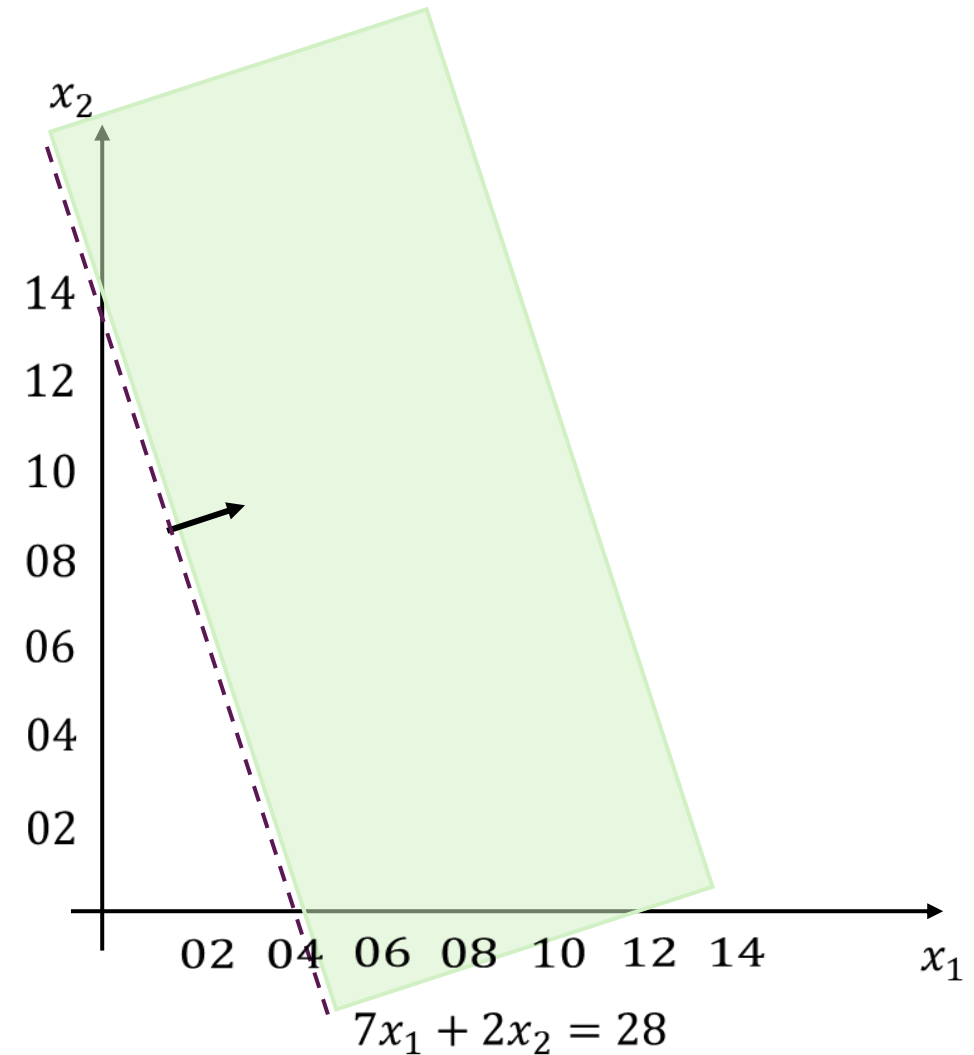


Example | Graphical Solution Method

Let's try to solve this LP:

$$\begin{array}{ll}\min z = & 50x_1 + 100x_2 \\ \text{s.t.} & 7x_1 + 2x_2 \geq 28 \\ & 2x_1 + 12x_2 \geq 24 \\ & x_1, x_2 \geq 0\end{array}$$

Step 1: Find $7x_1 + 2x_2 \geq 28$



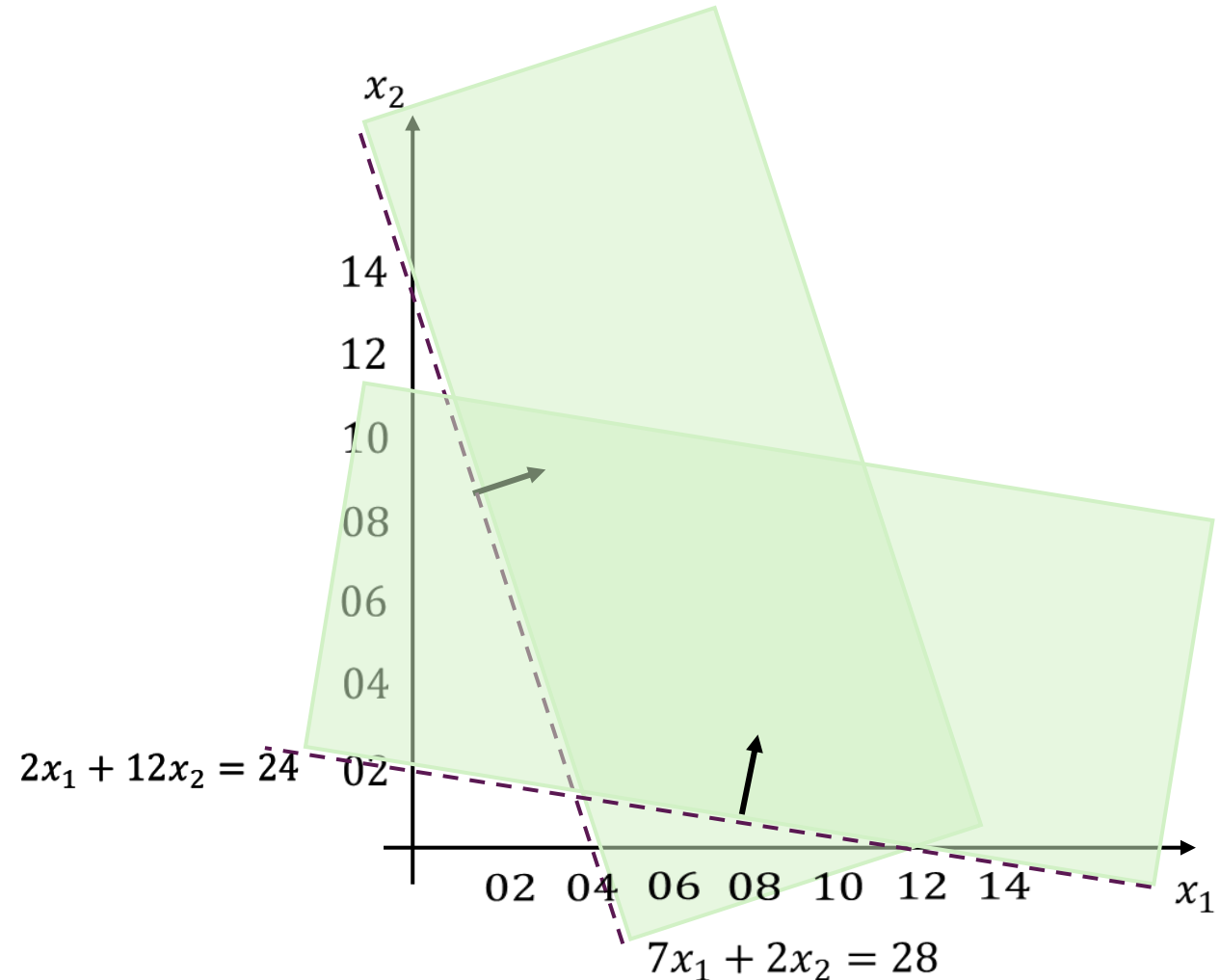
Example | Graphical Solution Method

Let's try to solve this LP:

$$\begin{array}{ll}\min z = & 50x_1 + 100x_2 \\ \text{s.t.} & 7x_1 + 2x_2 \geq 28 \\ & 2x_1 + 12x_2 \geq 24 \\ & x_1, x_2 \geq 0\end{array}$$

Step 1: Find $7x_1 + 2x_2 \geq 28$

Step 2: Find $2x_1 + 12x_2 \geq 24$



Example | Graphical Solution Method

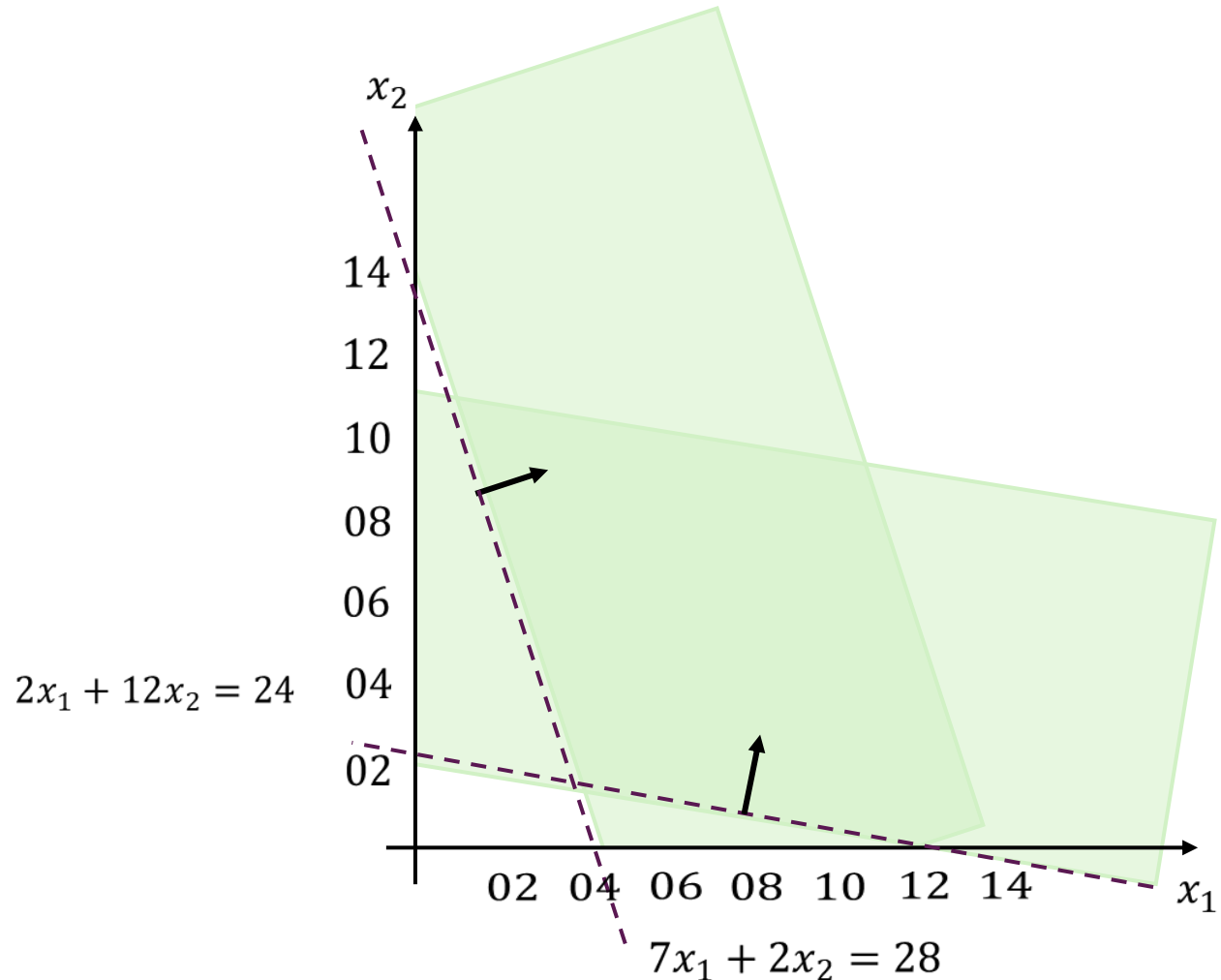
Let's try to solve this LP:

$$\begin{array}{ll}\min z = & 50x_1 + 100x_2 \\ \text{s.t.} & 7x_1 + 2x_2 \geq 28 \\ & 2x_1 + 12x_2 \geq 24 \\ & x_1, x_2 \geq 0\end{array}$$

Step 1: Find $7x_1 + 2x_2 \geq 28$

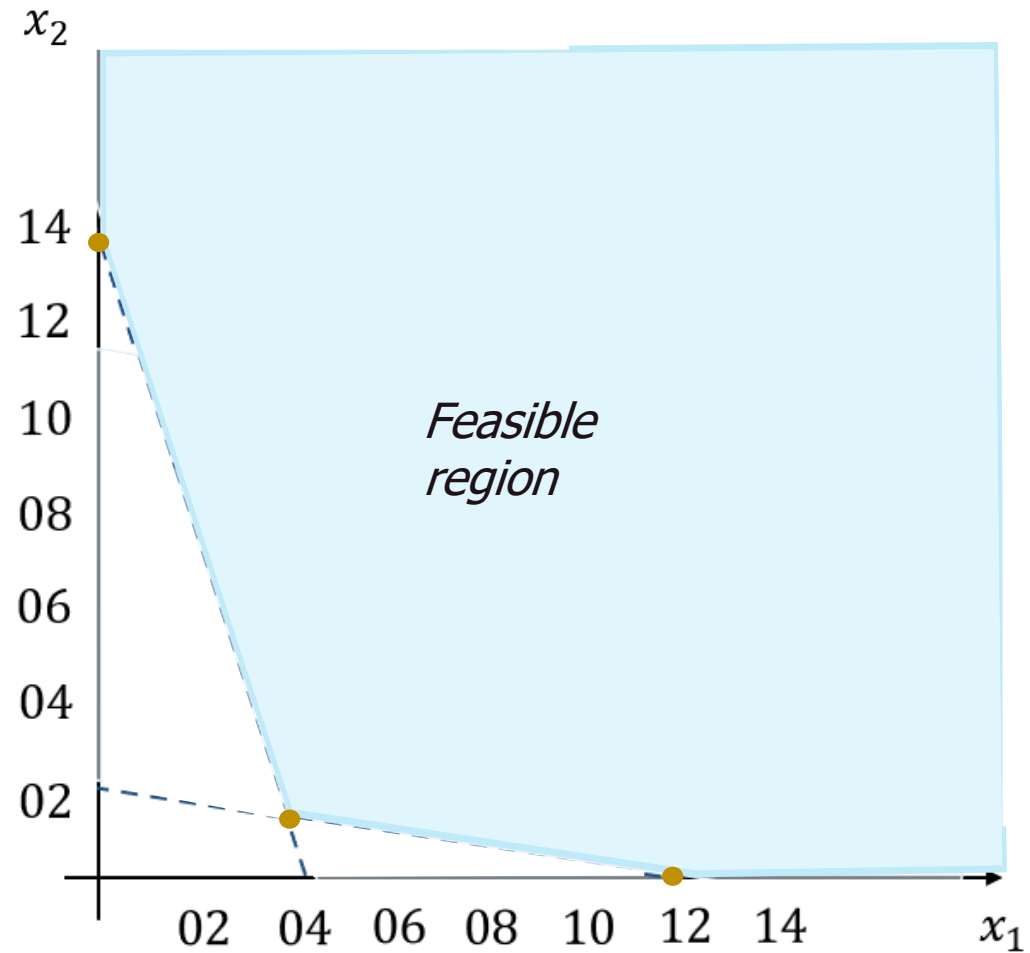
Step 2: Find $2x_1 + 12x_2 \geq 24$

Step 3: Get rid of $x_1, x_2 < 0$



Example | Graphical Solution Method

Step 4: Define the feasible region

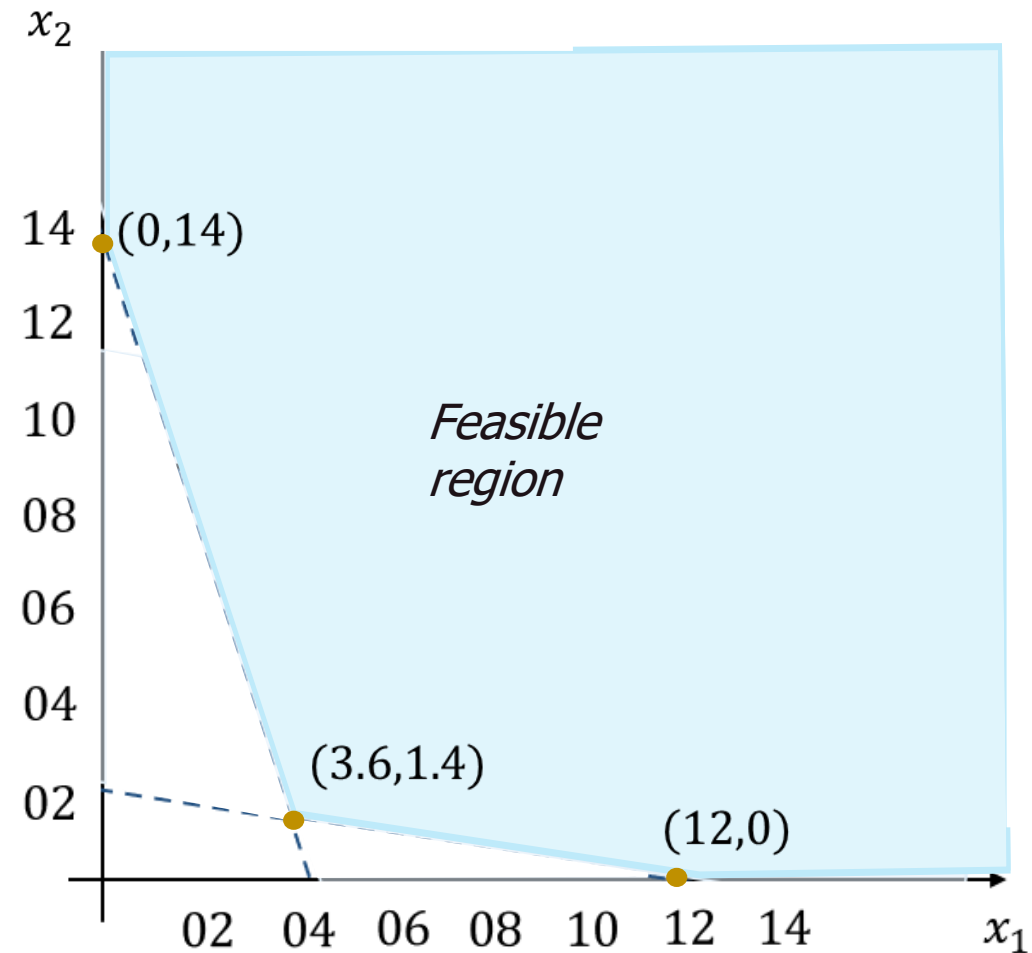


Example | Graphical Solution Method

Step 4: Define the feasible region

Step 5: Find corner points (*if finite*)

$$\left. \begin{array}{l} \text{via } 2x_1 + 12x_2 = 24 \\ \quad 7x_1 + 2x_2 = 28 \end{array} \right\} \text{intersection}$$
$$x_1 = 3.6, x_2 = 1.4$$



Example | Graphical Solution Method

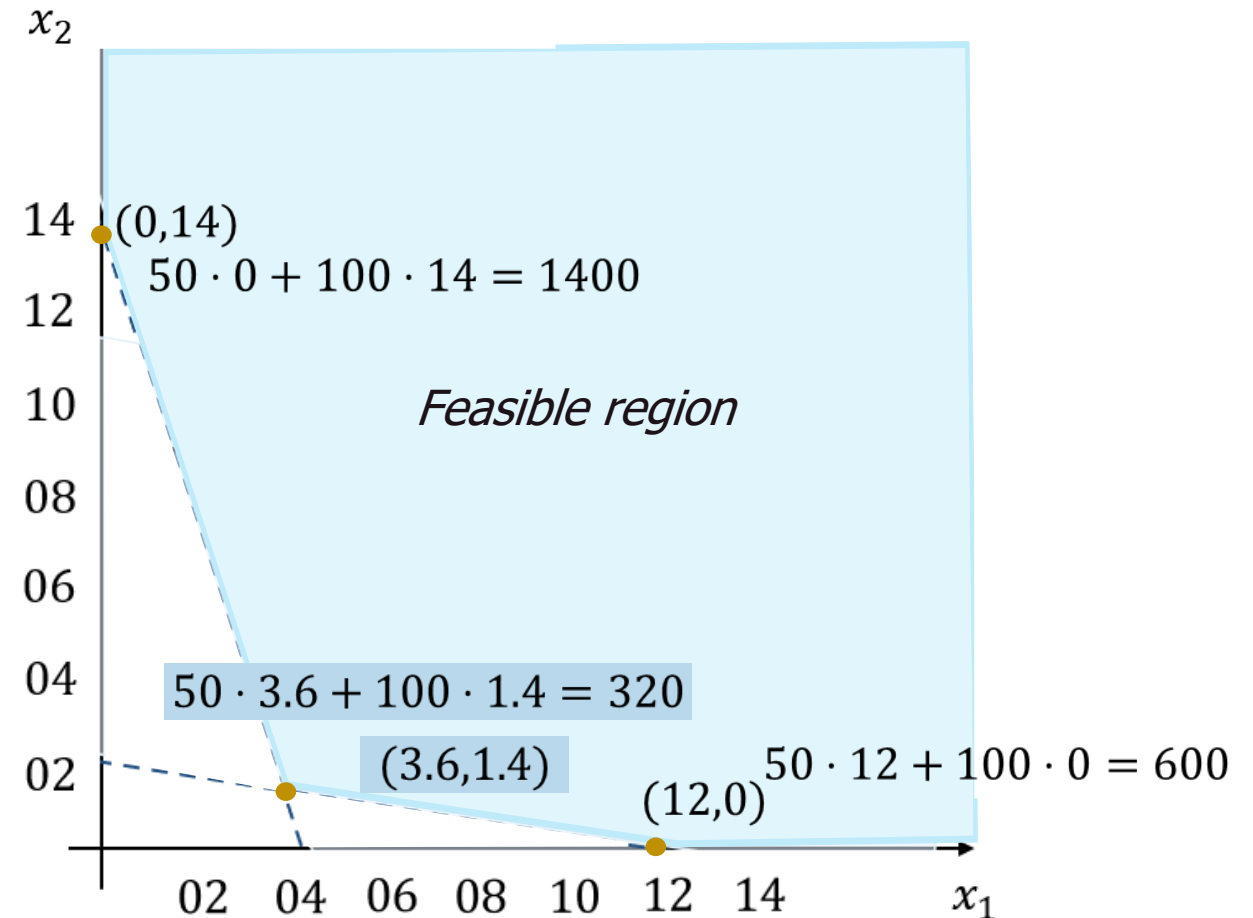
Step 4: Define the feasible region

Step 5: Find corner points *(if finite)*

$$\left. \begin{array}{l} \text{via } 2x_1 + 12x_2 = 24 \\ \quad 7x_1 + 2x_2 = 28 \end{array} \right\} \text{intersection}$$
$$x_1 = 3.6, x_2 = 1.4$$

Step 6: Compare corner point values and pick the optimal!

Note: Objective function is to minimize z and not maximize!

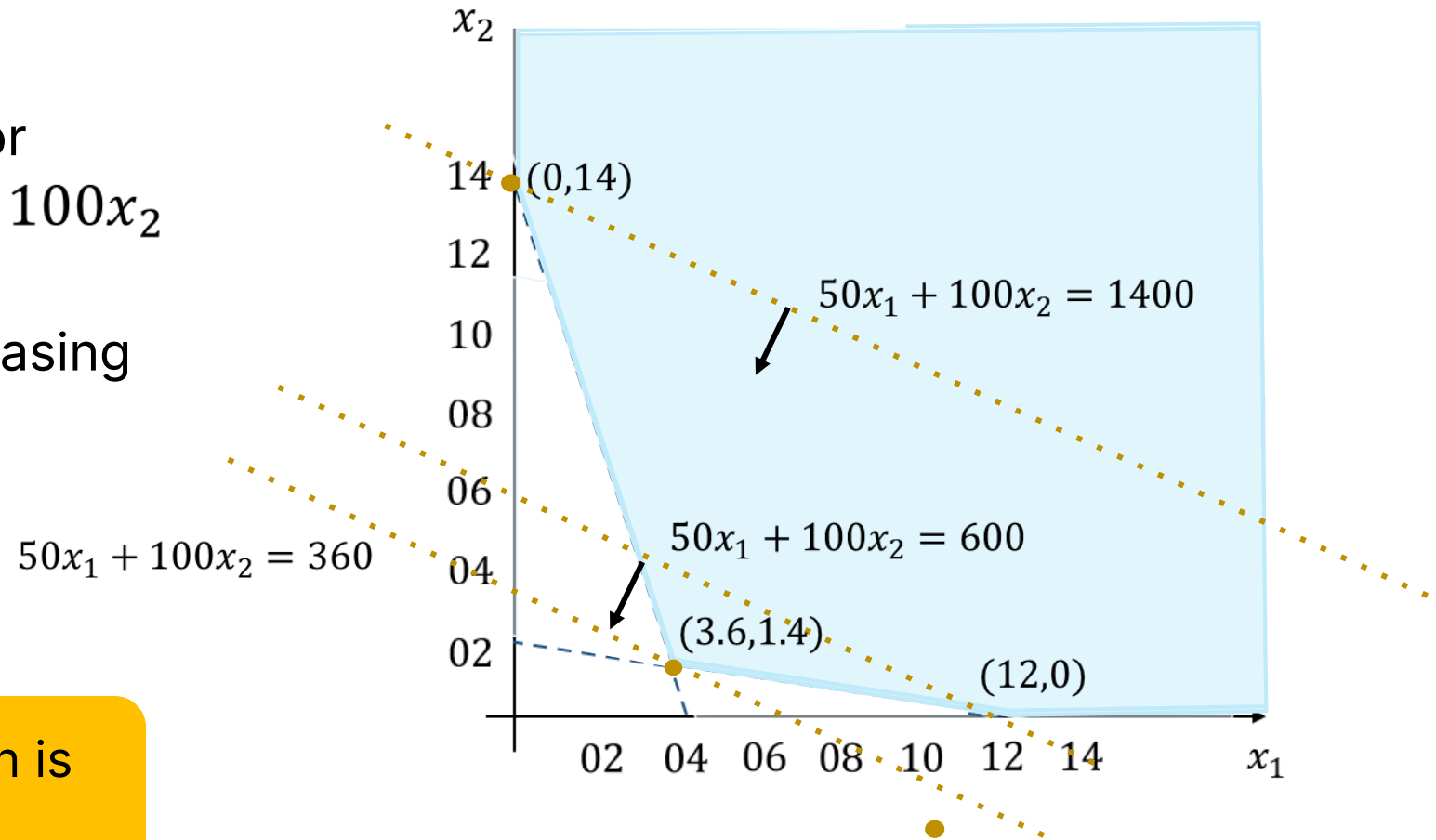


Example | Graphical Solution Method

Different approach:

- Draw iso-value line for
 $\min z = 50x_1 + 100x_2$
- Slide the line in decreasing direction (minimization problem)

The optimal solution is
 $x_1, x_2 = (3.6, 1.4)$



Reading Suggestion:

Extreme Points and Basic Feasible Solutions

Section 7.1 from Veatch's Book, pg 134-136